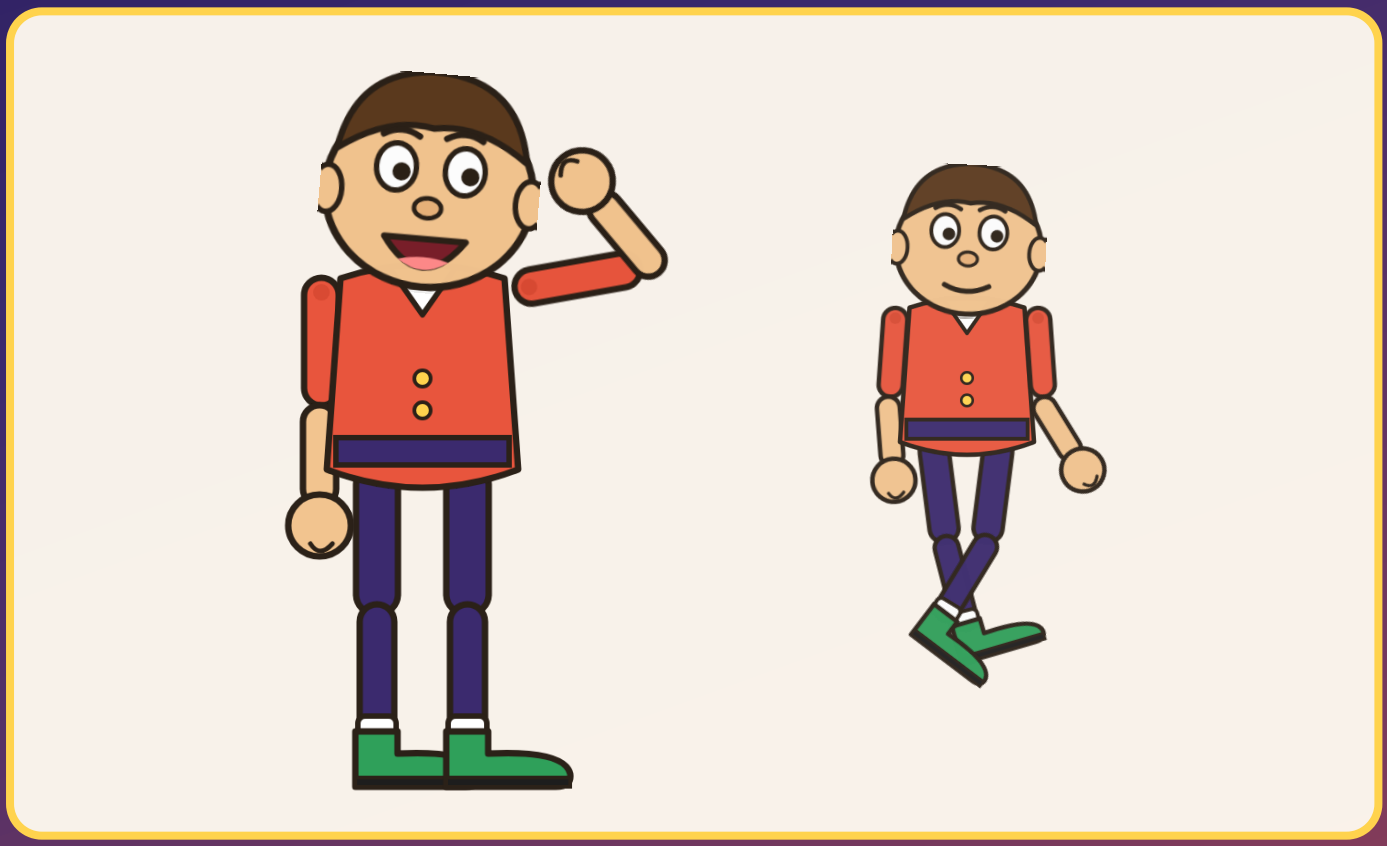
H-B TOONSUITE

H-B ToonSuite

Professional 2D Animation Software — Complete User Manual



From the history of limited animation to your first finished cartoon

Manual for H-B ToonSuite v5 · English edition

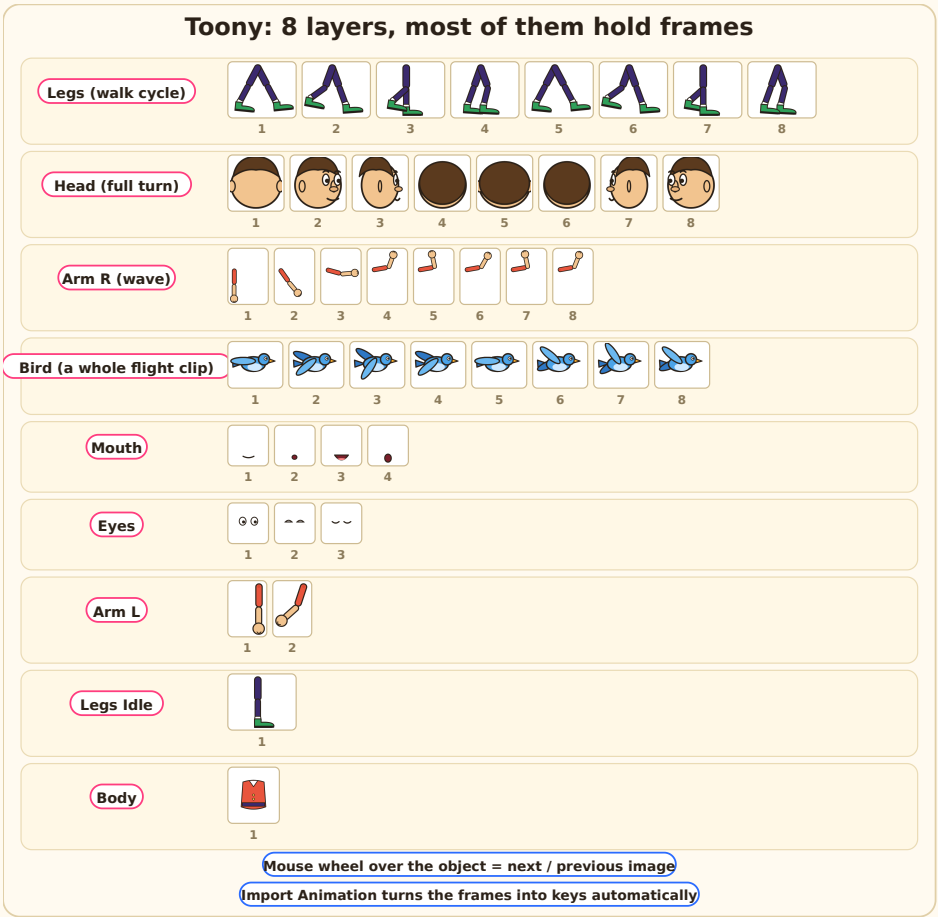
Contents

1	Welcome to H-B ToonSuite	3
2	History: Hanna, Barbera and Limited Animation	6
3	Purpose and Workflow	9
4	First Launch and a Tour of the Screen	11
5	The Menus	13
6	The Toolbar, Tool by Tool	18
7	Staging: Layers, Folders and Images	21
8	Dropping on the Stage and the Rig Page	29
9	Transform, Pivot, Link and Offset	34
10	Deforming: Clay, Bend, Stretch, Squash, Skew, Hide, Flip	40
11	Keys and the Timeline	46
12	The Graph Editor	50
13	The Camera and Effects	52
14	Reusing Work: Clips and Characters	55
15	Export, Settings and Performance	56
16	Anatomy of a Frame-Based Character	60
17	Complete Project: “Toony Says Hello”	65
18	Tips, Troubleshooting and Cheat Sheet	76
A	Appendix: Default Keyboard Shortcuts	78

1 Welcome to H-B ToonSuite

H-B ToonSuite is a complete 2D animation studio that lives inside a single HTML file. Its heart is one simple but powerful idea: **a layer can carry a whole sequence of frames inside itself.** A layer can hold the eight drawings of a walking pair of legs, the eight views of a head turning all the way round, six drawings of a waving hand – or even a complete little motion clip. You place that layer on the stage once, and the program plays the frames inside it exactly when you want.

> **Note**
>
> This program is **not a cut-out (paper-puppet) tool.** You do not have to build a character from bones and swing every limb with rotation keys. In ToonSuite the animation lives in the **frames you draw**; transforms, links, the camera and the deformers (Clay, Bend, Stretch, Squash, Skew) are there to **place, match, polish and enrich** those frames.



The big idea: one layer = one little film. Most layers of a character hold several frames.

What you can do with it

- Store **cycles and clips inside layers**: a walk, a run, a head turn, a hand wave, a bird's flight – and replay them with **Import Animation** (repeat, reverse, hold).
- Swap faces: eyes and mouths are small layers with several images – turn the **mouse wheel** to change them, every swap is keyed.
- Group layers in **folders** (a folder is a character, a prop or a whole scene) and move the folder as one object.
- **Link** layers (a face follows a head), set **pivots**, and animate position, rotation and scale with keys.
- Use **Clay, Bend, Stretch, Squash and Skew** to match layers to each other, to give figures a lovely hand-made shape and to add squash-and-stretch.
- Polish timing in a full **Graph Editor**, film with a virtual **camera**, and export **MP4, WebM, JPG or PNG sequences**.

What you need

- A recent desktop browser (Chrome, Edge or similar) and a mouse with a middle button for panning.
- Your frames as **transparent PNG** files (or SVG / Lottie – see the Import chapter). Solid-colour backgrounds can be removed automatically.
- A good habit: every frame of one layer has the **same canvas size** and the part is drawn in the same place on it, so the frames line up perfectly when they change.

> **Note**
>
> The program interface is **English only and left-to-right**. All button names in this manual are written exactly as they appear on screen, in a special style like **Auto Key**, and keyboard shortcuts look like `Ctrl+S`.

What is new in this edition (version 7)

This manual covers version 7 of the program. The new or improved features are explained in the chapters listed here:

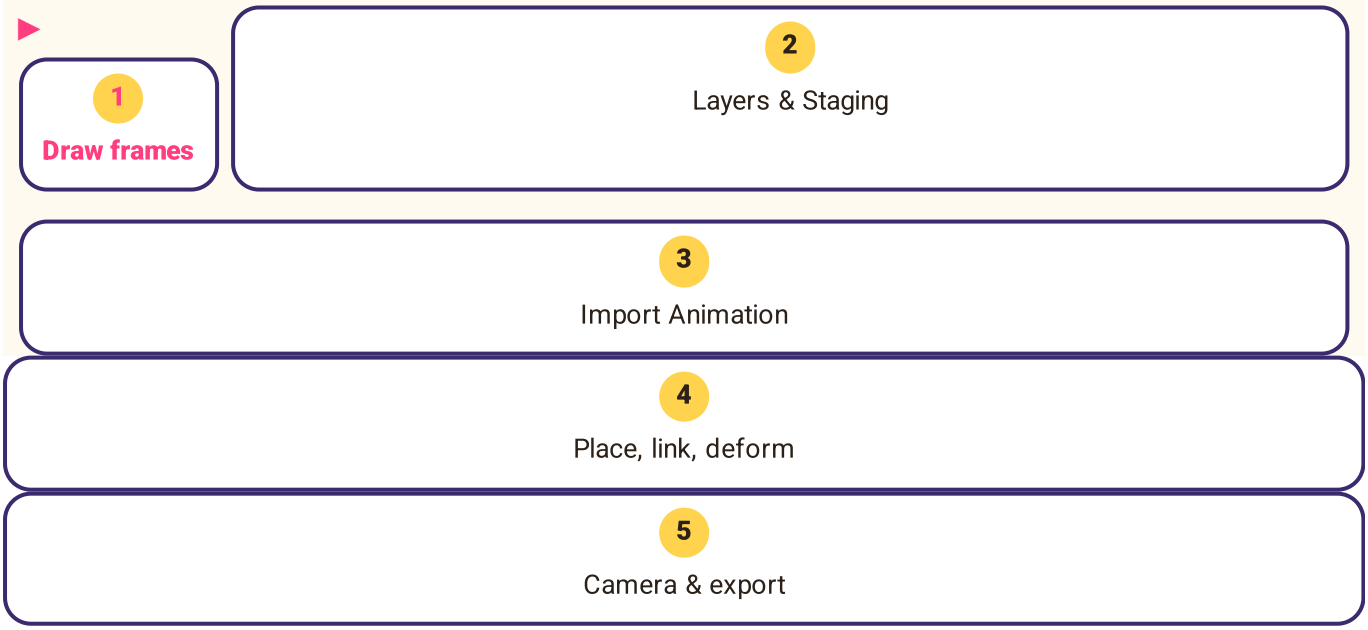
New / improved	Where
Clay is now a Liquify-style tool: twirl, pucker, bloat, mesh subdivision, brush-size by wheel, targeting overlapping layers with `Alt`+click, brush panel with reset.	Chapter 10
Rig page (Character Layout): bring layers to front / up / down / back, flip a single layer, **Spread** the parts, and Apply keeps the new order.	Chapter 8

New / improved	Where
Staging rows have Lock, Eye, Freeze and Solo icons; a **Spread** button lifts layers out of a pile so a hidden one can be picked.	Chapter 7
Image counter (for example “2 / 8”) appears when you step through the frames of a layer; a big **frame number** flashes while you scrub the timeline.	Chapters 7, 11
Camera: Camera View, **Gizmo** (camera schematic), colour and opacity of the **Outside** area, and a High / Preview quality for Clay and Bend.	Chapter 13
Many fixes: undo / redo, deleting scene objects, layer order and flips kept when you apply the Rig page, and a more reliable timeline.	—

How to read this manual

The book follows the order in which you will really work: history and idea, a tour of the screen, then every tool in detail, then a **character chapter** with diagrams, and finally a **complete small project** from zero to a finished video. Pink numbered badges like (1) on a screenshot point to the item with the same number in the legend below it.

> **Tip**
>
> Beginner? Read chapters 1–4, then jump to chapter 17 and build the project with the book open beside you. Come back to the reference chapters when you want details.



2 History: Hanna, Barbera and Limited Animation

The name **H-B** in H-B ToonSuite is a tribute to **William Hanna and Joseph Barbera**, the animation pair who turned cartoons into a television industry. The program is built around the technique they made famous: **limited animation**.

1940 Hanna and Barbera, working at the MGM studio, create the cat-and-mouse pair Tom and Jerry – lavish, fully animated theatrical shorts.

1957 MGM closes its animation department. Hanna and Barbera found their own company, Hanna-Barbera Productions, and aim at a new customer: television.

1957–58 Early TV series such as Ruff and Reddy and then Huckleberry Hound show that a cheaper, smarter way of animating can still delight audiences.

1960 Primetime comes next: The Flintstones is widely cited as the first animated series in American prime time.

1960s–70s Dozens of characters follow – bears, dogs, cats, space-age families, mystery-solving teenagers – all produced with the same efficient method.

What is limited animation?

Classic Disney-style animation redraws almost everything on every frame – up to 24 drawings per second. For television, with weekly deadlines and small budgets, that was impossible. Hanna-Barbera's answer was to **draw less and design smarter**. The main ideas:

- **Redraw only what moves.** The character is designed in simple shapes; a collar, tie or scarf cleanly separates the head from the body. Only the moving part – a mouth, a head, a pair of legs – gets new drawings, while the rest of the figure stays on its cel.
- **A short series of drawings per action.** A walk is drawn as a handful of leg drawings; a head turn as a handful of head drawings; a wave as a handful of hand drawings. These small sets are the real "units" of the film.
- **Reuse in cycles.** Those drawings are repeated in a loop – a walk cycle, a run cycle, a flapping bird – and the background slides behind them.
- **Hold poses.** A character can stay still while only the mouth talks – the voice and the writing carry the scene.

- **Simple, reusable backgrounds** and strong, readable character design, with timing and sound doing the acting.

Why this idea fits a computer perfectly

A cel with a small set of drawings that you call up again and again is exactly what a **layer with several images** is. H-B ToonSuite turns each old studio habit into a feature:

Classic technique	In H-B ToonSuite
Walking legs drawn once, repeated in a cycle	A **Legs layer with 8 images**; **Import Animation** repeats it (Repeat, Frames per image, Load reversed).
A set of head drawings for a turn	A **Head layer with 8 views** from front to back and round again.
A series of hand drawings for a wave	An **Arm layer with the whole wave** inside it.
A whole action (a bird flapping) on one cel set	A complete **clip inside ONE layer**; replay it as many times as you like.
Mouth drawings swapped per sound	Several images on one layer – **mouse wheel** swaps them, each swap is keyed.
Holding a drawing for several frames (“on twos”, “on threes”)	**Frames per image** and stepped keys.
Painters matching one cel to the next	**Clay / Bend / Stretch / Squash / Skew** to match layers and shape figures.
The rostrum camera over the artwork	A virtual **Camera** with zoom, rotation and effects.

Reading a classic model sheet

Study the character sheets of classic limited-animation shows (a gang of alley cats is a perfect example): flat colours, a strong outline, round simple masses, a vest or scarf marking the neck. When you plan your own character, look for the same natural **seams** and ask three questions:

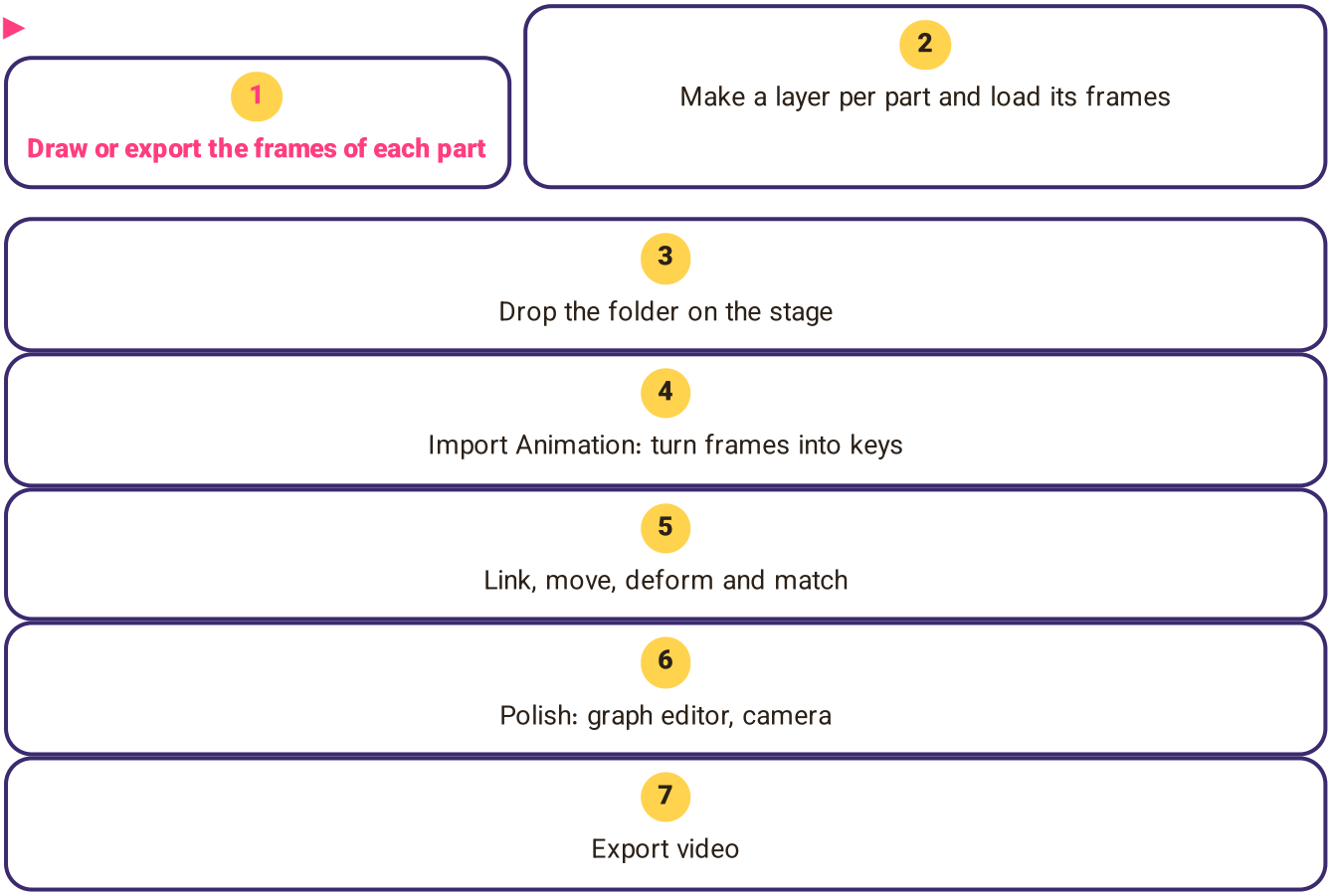
1. **What stays still?** (the torso, the vest) – it becomes ONE layer with ONE image.
2. **What moves in a repeating way?** (legs, a tail, wings) – it becomes a layer with a cycle of images.
3. **What changes expression?** (eyes, mouth, brows, a head turning) – it becomes a layer with a small family of images.

Note

Character and show names mentioned in this chapter are trademarks of their respective owners. They are named here only for historical context. All characters drawn in this manual (“Toony” and the little bird) are original artwork made for this book.

3 Purpose and Workflow

The goal of H-B ToonSuite is simple: **let one person make a good-looking TV-style cartoon quickly**, using the proven efficiency of limited animation and the comfort of a modern editor. Everything in the program serves one pipeline.



Two ways to animate – and you use both

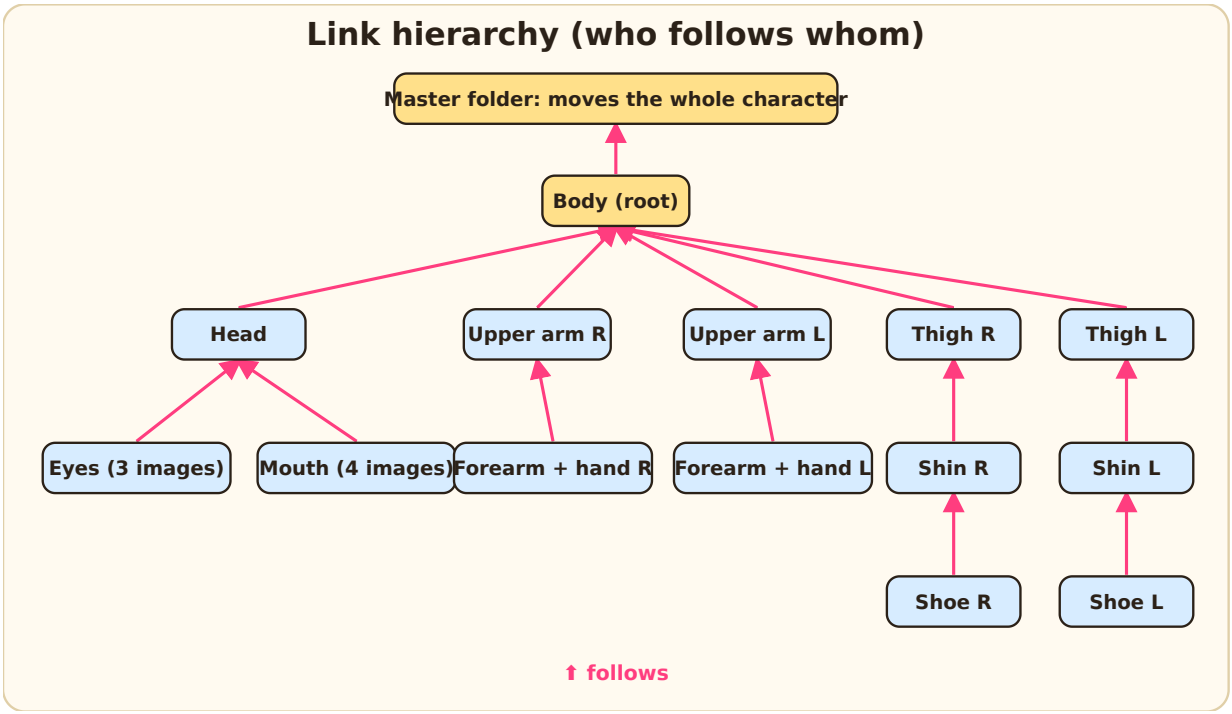
	Frames inside layers	Transform keys
What changes	Which image of the layer is shown	Position, rotation, scale, deformation
Typical use	Walk cycles, head turns, hand gestures, mouth and eye swaps, whole clips	Moving a character across the stage, camera moves, bouncing, squash and stretch
How it moves in time	Jumps from image to image (stepped keys)	Smooth curves you edit in the Graph Editor
Fastest way to create	**Import Animation**	**Auto Key** and the numeric bar

The two worlds: Staging and Stage

Place	What it is	You use it to…
Staging (right panel)	The warehouse: a tree of layers and folders with their images.	Create layers, load frames, group, reorder.
Stage (center)	The set: what the audience sees inside the white frame.	Place, move, link and animate objects.
Scene tab	A tree view of what is currently on the stage.	Inspect the hierarchy of linked parts.

A folder is a character (or a prop, or a scene)

When you drop a **folder** onto the stage, every layer inside it becomes a separate stage object, and the folder itself becomes a **master** object that moves them all together. The character is therefore ONE thing (it walks across the screen) made of a few layers, each of which plays its own little film.



The link tree of our sample character. The master folder carries everything; the face follows the head.

> **Tip**
>
> Because the legs, head and arm already contain their own motion, you rarely need long chains of links. Link only what must follow something else – a face follows a head, a hat follows a head.

4 First Launch and a Tour of the Screen

Open the **H-B_ToonSuite_v5.html** file in your browser. A welcome screen appears; tick `Do not show this screen again` if you do not want to see it each time, then click anywhere to enter the workspace. (The same option exists in Project Settings as “Show the intro screen when the program starts”.)

The H-B ToonSuite workspace with a finished project loaded.

1. **Menu bar** – File, Edit, View, Layers, Help.
2. **Toolbar** – tools for transforming, linking, deforming, camera and more.
3. **Stage** – the white frame (or the camera view) where the scene lives.
4. **View controls** – undo, redo, zoom percentage, `Reset View`.
5. **Side panel** – Staging and Scene tabs.
6. **Numeric bar** – exact X, Y, rotation and scale of the selection.
7. **Timeline** – frames, keys and the playhead.
8. **Animation range bar** – the first and last frame of the whole animation.
9. **Transport and key controls** – play, Set Key, Auto Key, Graph Editor, fps.

Moving around the stage

Action	How
Zoom the stage view	`Alt+Wheel`
Pan the stage view	Middle-mouse drag
Return to the default view	**Reset View** or View ▸ Reset Stage View
Zoom the whole program	`Ctrl+=` `Ctrl+-` `Ctrl+0`
Fullscreen	`F11`

> **Note**
>
> Browser zoom is blocked on this page, so `Ctrl+Wheel` and `Ctrl+-` keep working inside the app. Use the app-zoom shortcuts if the interface is too small on your monitor.

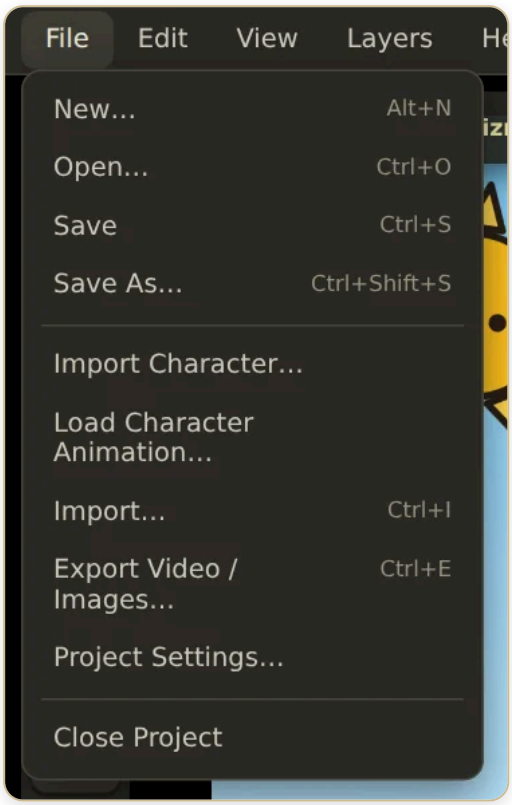
Saving your work

Use **File ▸ Save** `Ctrl+S` and **File ▸ Save As…** `Ctrl+Shift+S`. A project is saved as a file with all layers, images, rig, keys and cameras inside it. The option **Save Plus (numbered versions)** in Project Settings keeps a new numbered copy every time, which is a cheap insurance against mistakes. **Open…** `Ctrl+O` loads a project and **Close Project** returns to an empty workspace.

5 The Menus

Five menus hold everything that is not a tool. Shortcuts are shown at the right of each item.

File

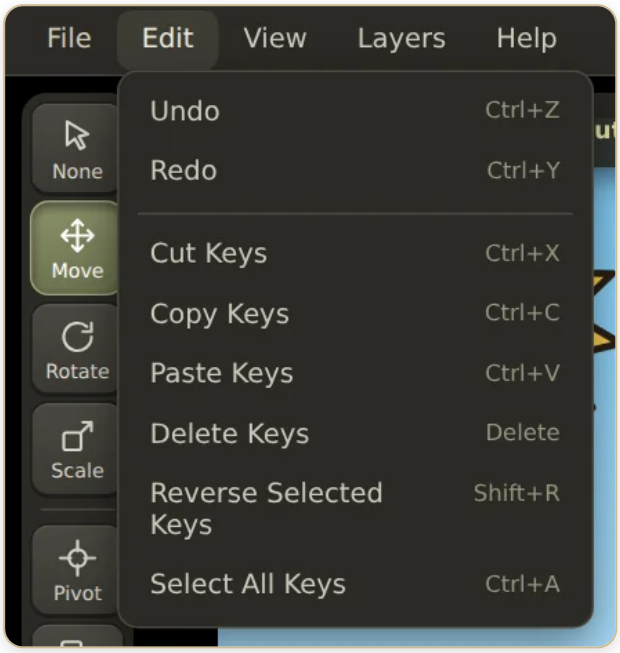


The File menu.

Item	What it does
New... `Alt+N`	Start a new empty project.
Open... / **Save** / **Save As...**	Load or store the project file.
Import Character...	Bring in a saved character (folder + layers + rig) made with “Save Character”.
Load Character Animation...	Load a character together with its animation (made with “Export Character Animation”).
Import... `Ctrl+I`	Import SVG vector art or Lottie animation as layers.
Export Video / Images... `Ctrl+E`	Open the export dialog (chapter 15).

Item	What it does
Project Settings...	Frame size, fps, range, defaults.
Close Project	Close the current project.

Edit



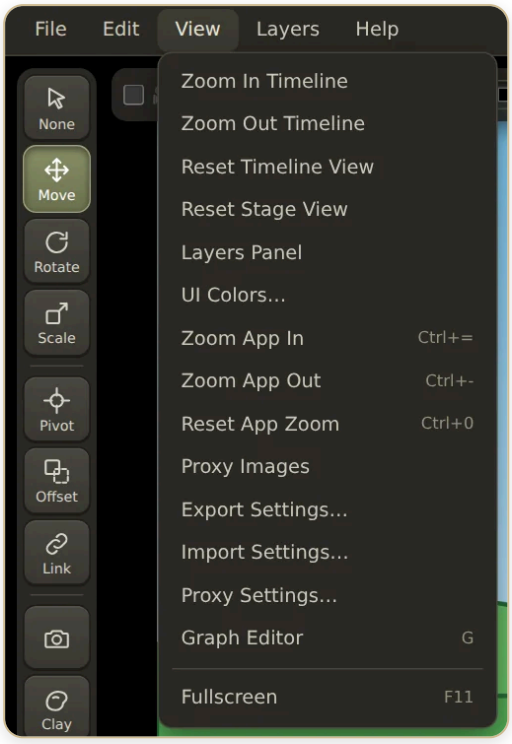
The Edit menu.

Undo/Redo work on your edits `Ctrl+Z), (Ctrl+Y`. The key items work on the selected keys of the timeline: **Cut, Copy, Paste**

`Ctrl+X/C/V) while the mouse is over the timeline), Delete Keys, Reverse Selected Keys ((Shift+R`

to play a motion backwards, and **Select All Keys** `Ctrl+A`.

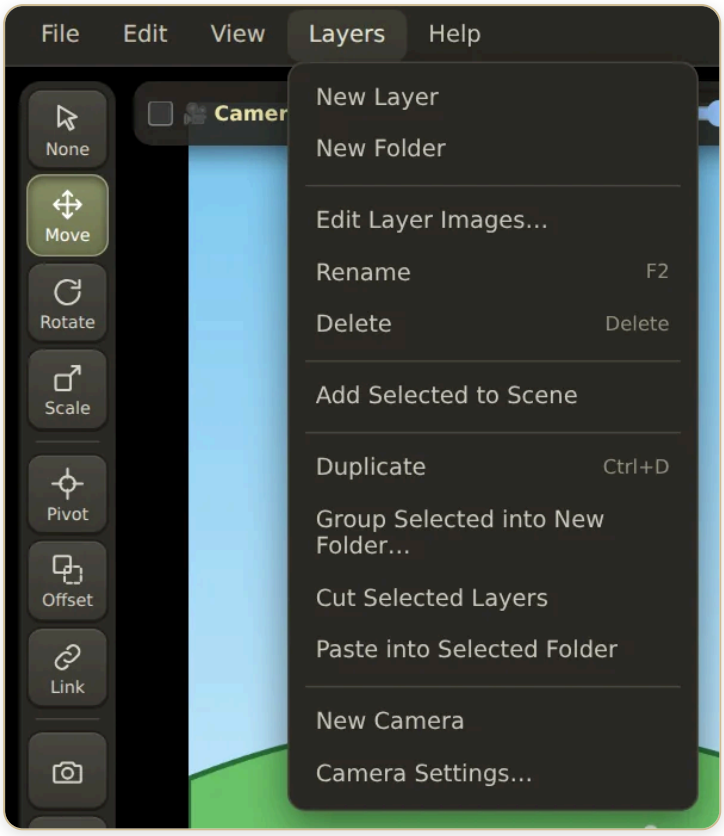
View



The View menu.

Item	What it does
Zoom In / Out Timeline, Reset Timeline View	Change how many frames are visible on the timeline.
Reset Stage View	Fit the stage into the window again.
Layers Panel	Show or hide the Staging/Scene panel on the right.
UI Colors…	Open the colour settings (chapter 15).
Zoom App In/Out, Reset App Zoom	Scale the whole interface.
Proxy Images, Proxy Settings…	Faster playback with scaled-down copies (chapter 15).
Export / Import Settings…	Save or load your colours, zoom and shortcuts to a file.
Graph Editor `G`, Fullscreen `F11`	Open the curve editor / go fullscreen.

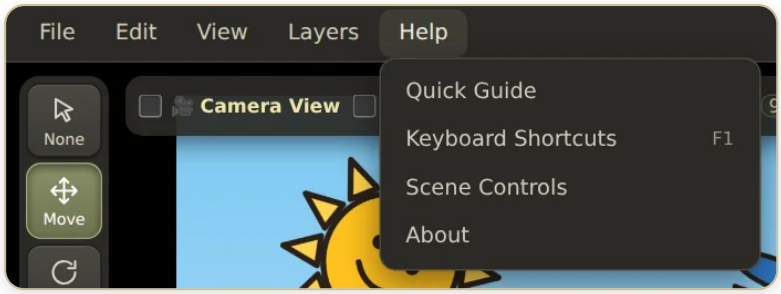
Layers



The Layers menu.

Item	What it does
New Layer / New Folder	Add to the Staging tree.
Edit Layer Images…	Open the image dialog for the selected layer.
Rename `F2` / Delete	Rename or remove the selection.
Add Selected to Scene	Put the selected layers or folder on the stage (same as dragging).
Duplicate `Ctrl+D`	Copy the selected layer or folder.
Group Selected into New Folder…	Wrap several layers in a new folder.
Cut Selected Layers / Paste into Selected Folder	Move layers between folders.
New Camera / Camera Settings…	Create a camera or open its panel (chapter 13).

Help

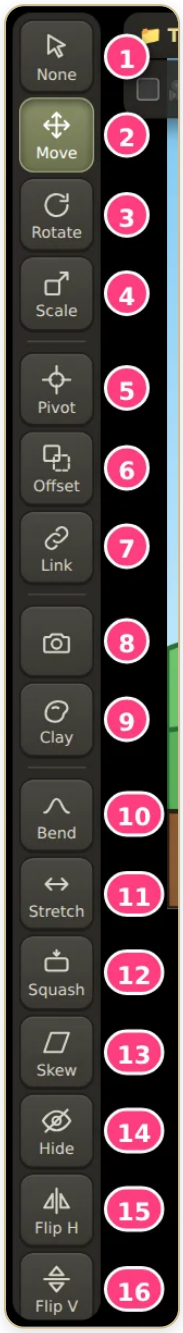


The Help menu.

Quick Guide is a compact built-in summary, **Keyboard Shortcuts** F1 opens the full, editable list of shortcuts and mouse gestures, **Scene Controls** explains mouse controls on the stage, and **About** shows the program information.

6 The Toolbar, Tool by Tool

The vertical toolbar on the left holds eighteen controls. Click one, or press its shortcut while the mouse is over the stage. **Right-click** a tool button to see its extra options.

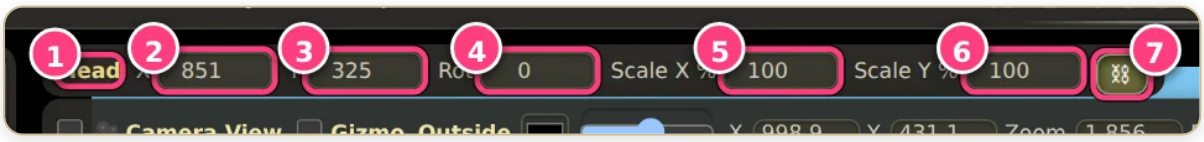


1. None Q
2. Move W
3. Rotate E
4. Scale R
5. Pivot P
6. Offset O
7. Link L
8. Camera C
9. Clay
10. Bend B
11. Stretch T
12. Squash U
13. Skew K
14. Hide
15. Flip H
16. Flip V
17. Global / Local
18. Reset Offset

The toolbar with numbered tools (top to bottom).

Tool	What it does
None	No active tool – you can only select objects. Safe mode for clicking around.
Move	Shows a red (X) and a green (Y) arrow plus a centre handle. Drag an arrow to move along one axis, the centre to move freely.

Tool	What it does
Rotate	Drag the ring around the pivot. Hold `Shift` to snap in 15° steps.
Scale	Resize from the pivot. The chain button in the numeric bar locks X and Y together.
Pivot	Moves the centre that rotation and scaling use – the "joint". Essential for elbows, knees, necks.
Offset	Adjust a pose on top of existing animation without creating keys (when Auto Key is off). Clear it with Reset Offset or bake it into keys.
Link	Click the child, then click the parent. From then on the child follows the parent. `Esc` cancels a pending link.
Camera	Turns the camera tool on/off (chapter 13).
Clay	Sculpt a layer like dough with a soft brush – a Liquify-style tool for matching layers and shaping figures (chapter 10). Right-click for the brush panel.
Bend Stretch Squash Skew	Drag over an object to bend it, stretch it long, squash it flat or slant it – for squash-and-stretch and for fitting layers together. See chapter 10.
Hide	Hides or shows the selected layer or folder. A hidden folder hides all its layers. Animatable.
Flip H Flip V	Mirror the selection horizontally or vertically. Keyed when Auto Key is on – perfect for turning a character around.
Global / Local	Switches the gizmo axes between the stage axes and the object's own (rotated) axes.
Reset Offset	Clears offsets on the selection (or on everything if nothing is selected).



The numeric bar appears above the stage when something is selected.

1. Name of the selected object.
2. X position (pixels in the frame).
3. Y position.
4. Rotation in degrees.
5. Scale X in percent.
6. Scale Y in percent.
7. Chain: lock X and Y scale together.

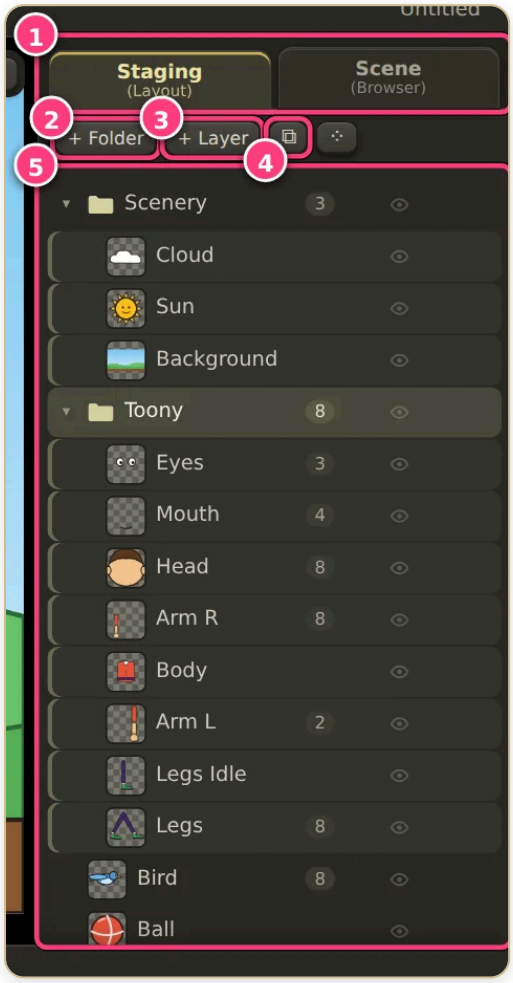
> **Tip**
>
> For exact values, type numbers in the numeric bar instead of dragging. With Auto Key on, the typed value is keyed on the current frame.

Tip

Nudge the selection with the arrow keys: 1 px, or 10 px with `Shift`.

7 Staging: Layers, Folders and Images

Staging is your warehouse. Everything you want to put on the stage is first built here as a tree of **layers** (a layer holds one or more images) and **folders** (a folder holds layers and other folders).



1. Tabs: **Staging** (Layout) and **Scene** (Browser).
2. `+ Folder` – new folder.
3. `+ Layer` – new layer.
4. Detach the panel into a floating window.
5. The layer tree: thumbnail, name, image count, eye icon to show/hide.

The Staging tab.

Creating layers and folders

Make your first folder

1. Click `+ Folder`, type a name such as **Toony** and press `Enter`.
2. With the folder selected click `+ Layer`, name it (for example **Body**) and press `Enter`. The layer is created inside the folder.
3. **Double-click** the layer to open the image dialog and add its picture(s).
4. Repeat for every part. The layer order in the tree is the front-to-back order: **top of the list is in front**.

The layer images dialog

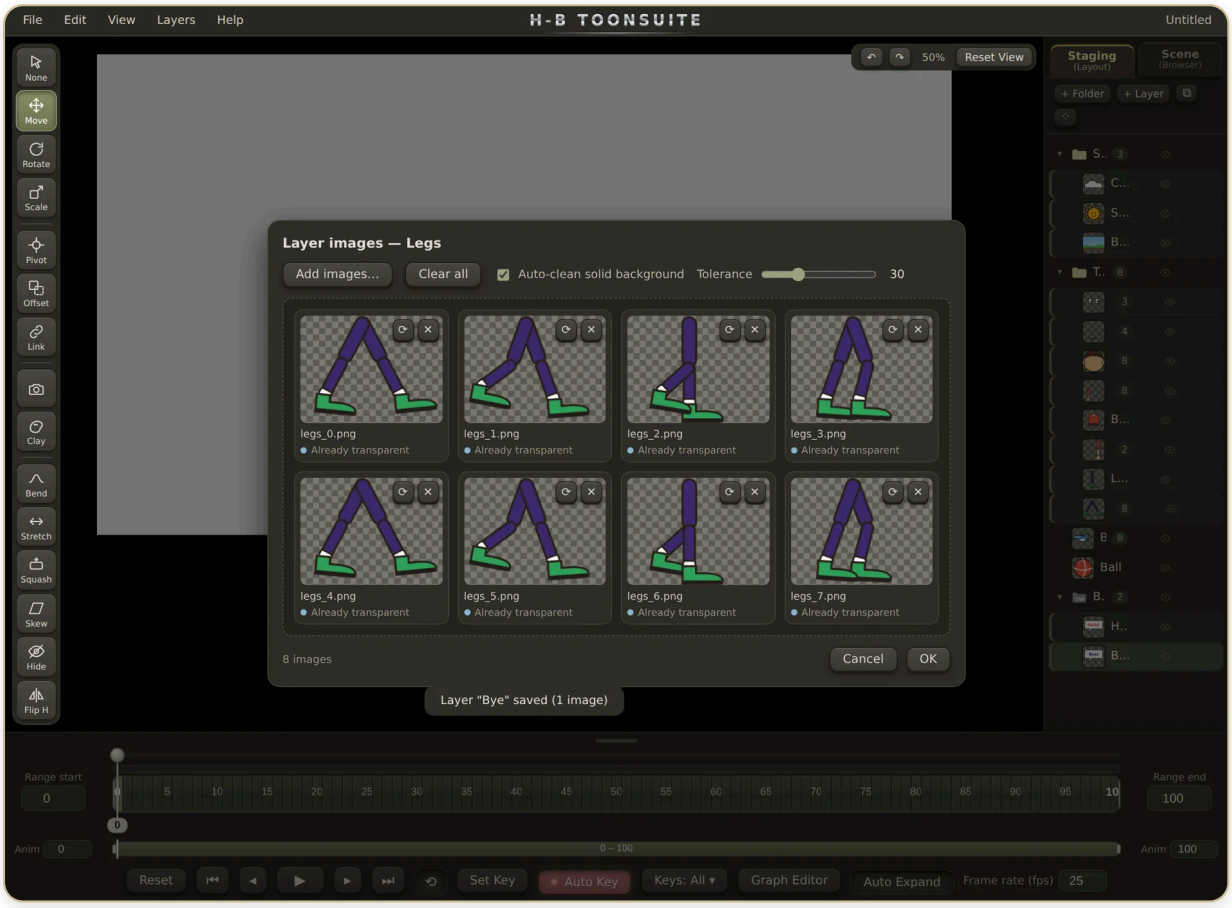
Layer images dialog for the Eyes layer, holding three images.

Control	Meaning
Add images…	Pick one or many PNG/JPG files. The images become the “slides” of the layer.
Clear all	Remove every image from the layer.
Auto-clean solid background	Turns a solid-colour background (for example white) transparent automatically.
Tolerance	How different from the background colour a pixel may be and still be removed (default 30).
Thumbnail buttons	⟳ replaces an image, × removes it. The caption shows “Already transparent” when no cleaning was needed.
OK / Cancel	Save or discard changes.

> **Tip**
> Use PNG files that already have transparency. Then the Auto-clean option has nothing to do and your edges stay perfect.

Layers that hold frames

Here is the feature that gives the program its name and soul. Open the image dialog of a layer and load **not one picture but a whole sequence** – the eight drawings of a walking pair of legs, for instance. All frames must share the same canvas size, so that the part never jumps when the picture changes.



The Legs layer holds eight walking frames. Each thumbnail is one drawing; ⟳ replaces it, × removes it.

Load a walk cycle into ONE layer

1. Draw (or export) the leg frames of one walking cycle as PNG files with the same canvas size: legs_1.png … legs_8.png.
2. Create a layer named **Legs** in the character folder and double-click it (or use `Layers ▸ Edit Layer Images…`).
3. Click `Add images…` and select all eight files together. Their order in the dialog is the playing order – drag nothing, just name them in order.
4. Press `OK`. The layer thumbnail shows the number of frames (here **8**).

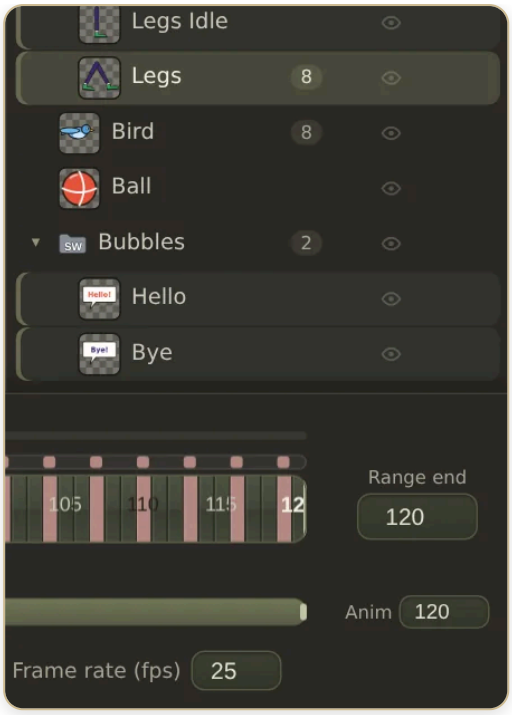
Playing the frames: Import Animation

Select the layer on the stage and press **Import Animation…** at the bottom of the side panel (the button also works for a whole folder – every layer with more than one image is converted). The program writes a stepped key for each image, so a whole cycle appears on the timeline in one click.

The Import Animation dialog, set up for the walking legs: 2 repeats, 3 frames per image, starting at frame 0.

Field	Meaning
Images in the layer	How many frames the layer holds (read-only).
Repeat (times)	How many times the whole sequence is played one after another – this is your cycle.
Load reversed (e.g. walk backwards)	Play the frames from last to first: walk backwards, lower an arm that was raised, close a door that was opened.
Frames per image	How long each drawing is held. 1 = “on ones” (smooth), 2 = “on twos”, 3 = “on threes” (the classic TV rhythm).
Start at frame	The timeline frame where the first image appears.

The length of the result is simple to compute: **images × frames per image × repeats**. Eight leg frames, held for 3 frames, repeated twice = 48 frames – almost two seconds at 25 fps.



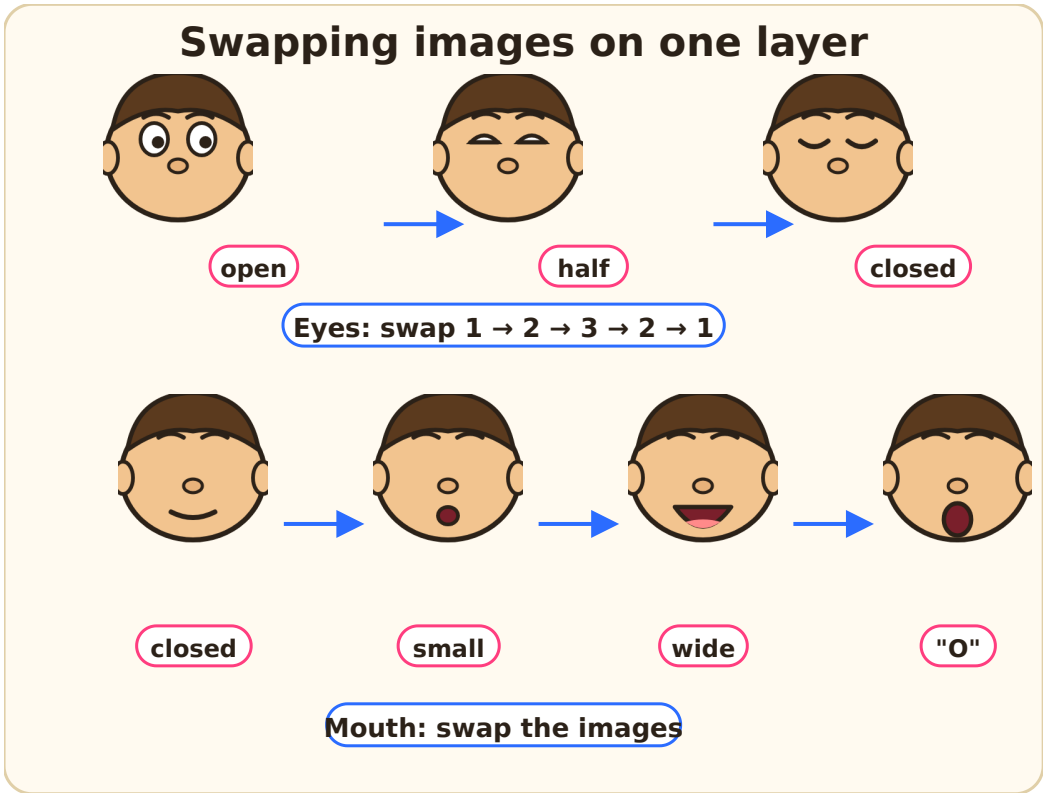
Bottom of the side panel: Reset, Freeze Transform and Import Animation.

> **Tip**
>
> Need a different rhythm in the middle of a scene? Just run Import Animation again for another time range – for example the same legs, reversed, to walk backwards at frame 98.

Stepping through frames by hand

On the stage, **scroll the mouse wheel over a layer** to show its next or previous frame. A counter such as **2 / 8** pops up so you always know which frame you are on, and with Auto Key on each change is keyed. This is how you set single poses, swap a mouth, or fix one frame of an imported cycle. Right-click the object and choose **Reorder Images…** to change the order of frames.

The counter “2 / 8” while stepping through the eight views of the head layer.



Small families of images: three eye states and four mouth shapes.

Row icons and the Spread button

Every row of the Staging tree carries small icons on its right side:

Icon	Meaning
Lock	Protects the layer or folder from accidental edits. Click again to unlock.

Icon	Meaning
Eye	Show or hide the layer or folder.
Freeze (snowflake)	The object stays visible but cannot be selected on the stage – ideal for a background you keep clicking by mistake. “Unfreeze this layer or folder to select it.”
S (Solo)	Temporarily hides every other layer so you can work on one. Click again: “Solo OFF: previous visibility restored”.

The small button next to the detach button is **Spread**: it lifts every layer out of the pile so you can pick a layer that is hidden behind others. Press Esc to leave Spread mode. The side panel itself is fairly narrow – long layer names are shortened, so name layers clearly from the start.

Layer and folder menus

Right-click a folder in Staging to open its menu:

Right-click menu of a folder in Staging.

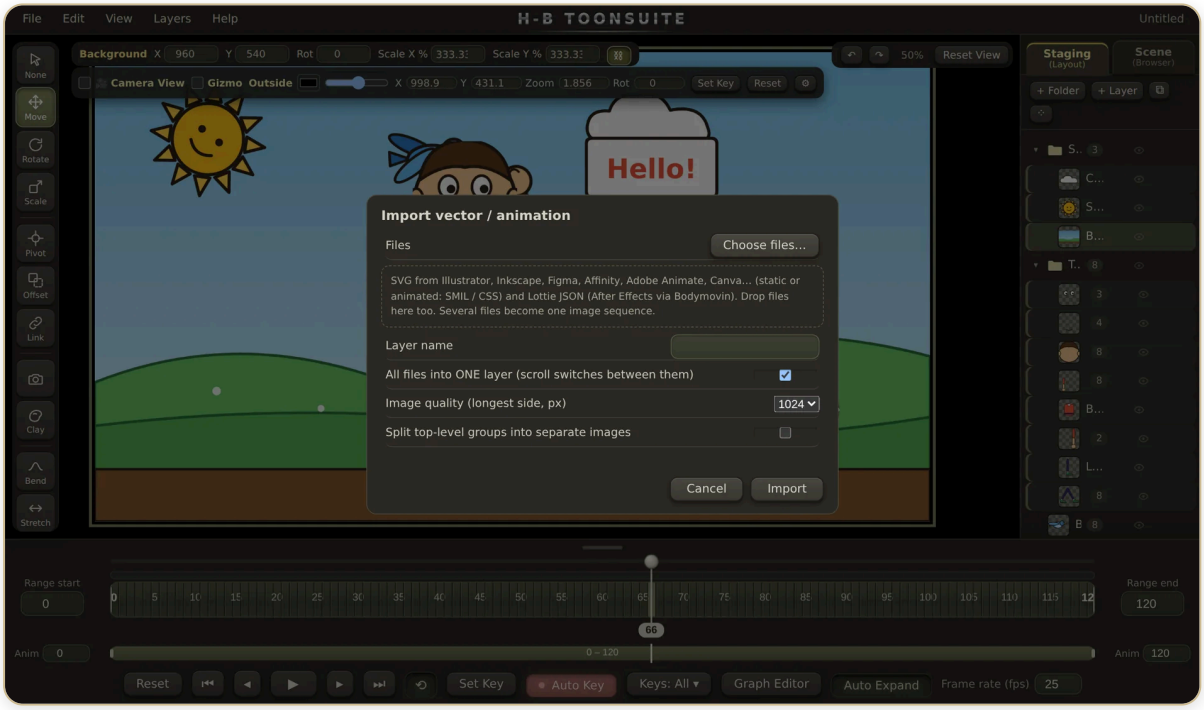
Item	Meaning
New Layer Here / New Folder Here	Add inside this folder.
Add to Scene	Put it on the stage.
Duplicate, Group into New Folder..., Cut, Move to “...”	Organise the tree.

Item	Meaning
Save Character (folder + layers + rig)…	Store this folder as a reusable character file.
Switch Folder	Turn the folder into a switch: only one child is visible at a time.
Split – choose the item to show…	For a switch folder: list the items and choose which one shows.
Rename `F2`, Delete Folder and its Layers…	Rename or delete.

Switch folders

A **switch folder** shows exactly one of its children. It is perfect for speech bubbles that change text, hand shapes, or alternate costumes. On the stage hold `Ctrl+Shift` and scroll over it to step through the items – the choice is animatable, so a bubble can say “Hello!” and then “Bye!”.

Importing vector art and Lottie



The Import dialog `Ctrl+I`.

File ▸ Import… reads SVG files from vector programs (Illustrator, Inkscape, Figma, Affinity, Adobe Animate, Canva…) and Lottie animations. Choose the files, set a **Layer name**, decide whether **all files go into ONE layer** (scroll the wheel to switch between them) and choose the **Image quality** (longest side in pixels, default 1024). **Split top-level groups into separate images** turns each top-level group of an SVG into its own image – a fast way to get a character’s parts from one drawing.

8 Dropping on the Stage and the Rig Page

Select a layer or folder in Staging and **drag it onto the frame** – or use `Layers ▸ Add Selected to Scene` or the Add to Scene item of the right-click menu.

A character folder placed on the stage. The message at the bottom explains what happened; here the scenery folder arrives small and is scaled up afterwards.

- Each layer becomes its own object. The folder becomes a **master** object that moves them together.
- Large artwork is **scaled down automatically** to fit comfortably. A big background folder arrives small – simply scale its master up (for example to 333 %).
- **Auto-link:** when a folder arrives, its biggest layer becomes the root and the other parts are linked to it. You then refine the links (see chapter 9).
- The wheel over an object steps through its frames; `Ctrl+Wheel` moves it in front of or behind the next layer.

Selecting on the stage

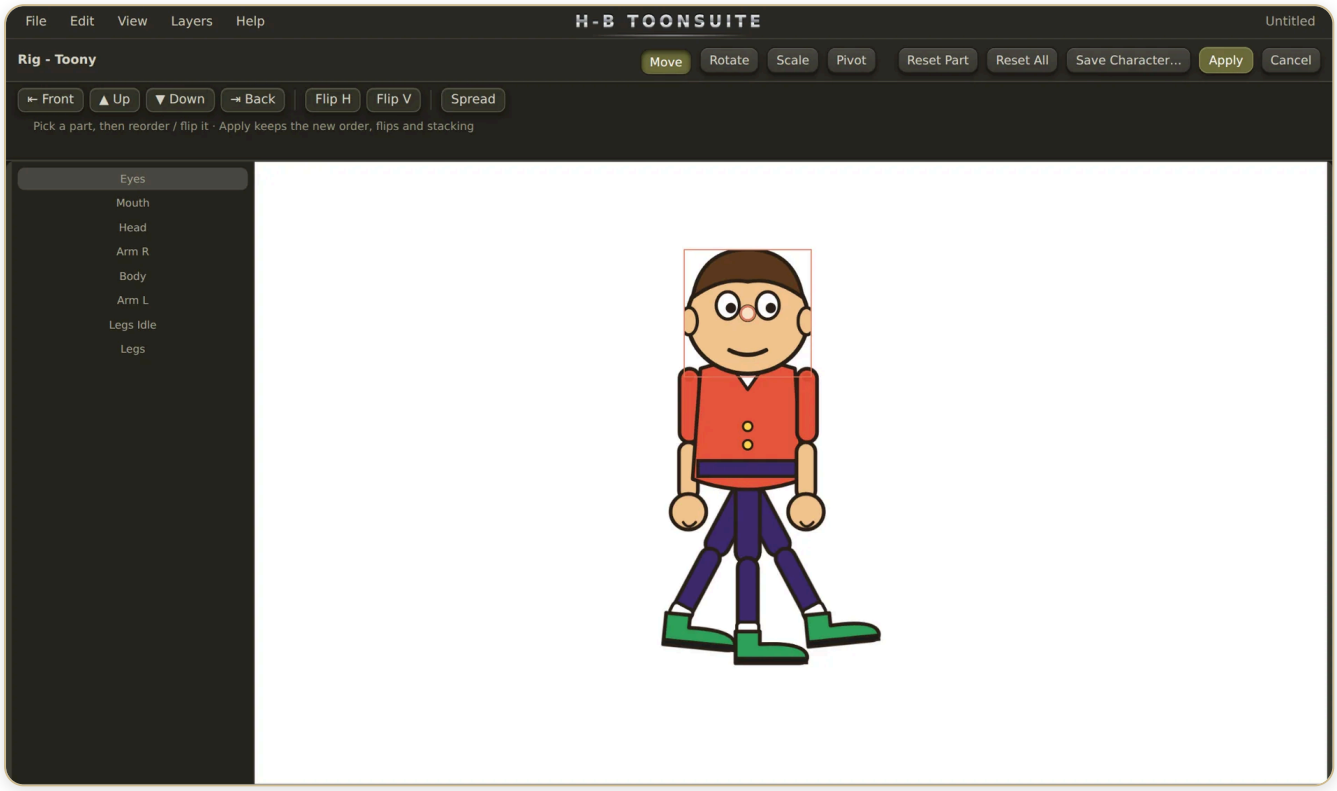
Action	Result
Click a folder component	Selects the whole folder (master). `Shift+click` selects just that one component.
`Shift` or `Ctrl`+click	Add / remove an object to the selection.
Drag on empty stage	Box-select objects.
Right-click an object	Opens the object menu (below).

Right-click menu of an object on the stage.

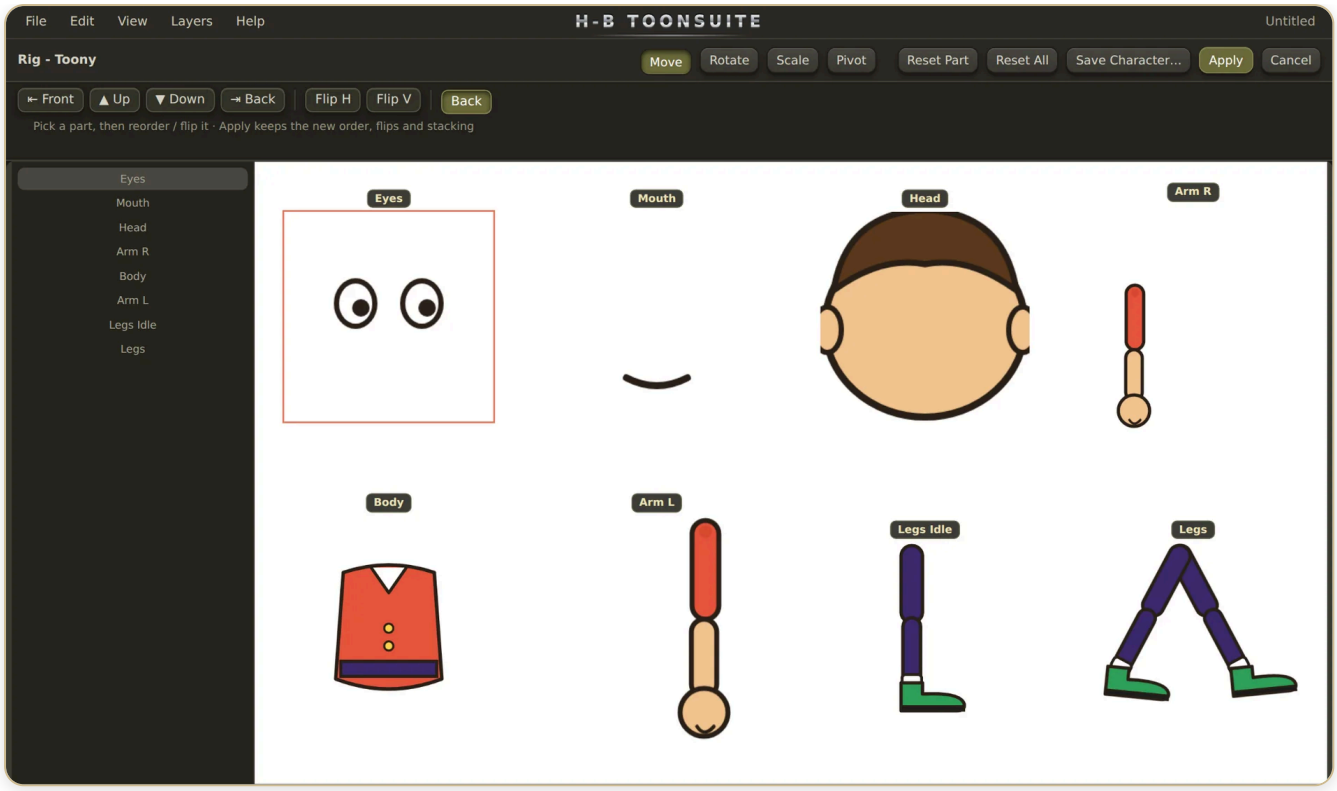
Item	Meaning
Find in Layers Panel	Highlights it in the Staging tree.
Export Clip (animation of this set)…	Save the keys of the selected set as a reusable clip.
Import Clip at this frame…	Paste a saved clip starting at the playhead.
Export Character Animation (character + animation)…	Save the character and its animation together.
Reorder Images… / Select Whole Folder	Change image order / select the master.
Reset Offset / Bake Offset into Keys	Clear or bake an offset (chapter 9).
Delete Object `Del`	Remove it from the stage.

The Rig page (Character Layout)

Double-click a folder to open the Rig page. It shows the character's layers on a white canvas so you can arrange the parts *before* you animate: where the head sits on the body, where the arms start. This is the "design the puppet" step.



The Rig page of the Toony folder. Notice the new row of order and flip buttons under the title bar.



Spread on the Rig page: every part of the character laid out with its name.

Control	Meaning
Move Rotate Scale Pivot	Choose what dragging a part does: place it, turn it, resize it, or set its pivot (joint).
Part list (left)	Click a name to pick a part, even if it is hidden behind another.
Front Up Down Back	Change the stacking order of the selected part: to the very front, one step up, one step down, to the very back. **Apply** keeps the new order.
Flip H Flip V	Mirror just the selected part.
Spread	Lays all parts out side by side with name labels – useful to pick, check and order them. Press Back to return.
Reset Part / Reset All	Undo layout changes of one part or of all.
Save Character...	Save the folder with its layers and rig as a character file.
Apply / Cancel	Keep or discard your layout.

Tip

Because the frames inside a layer already contain the motion, the Rig page is mostly about **where each layer sits and in which order**. Set a pivot only where something really rotates – for example at the neck if the head tilts.

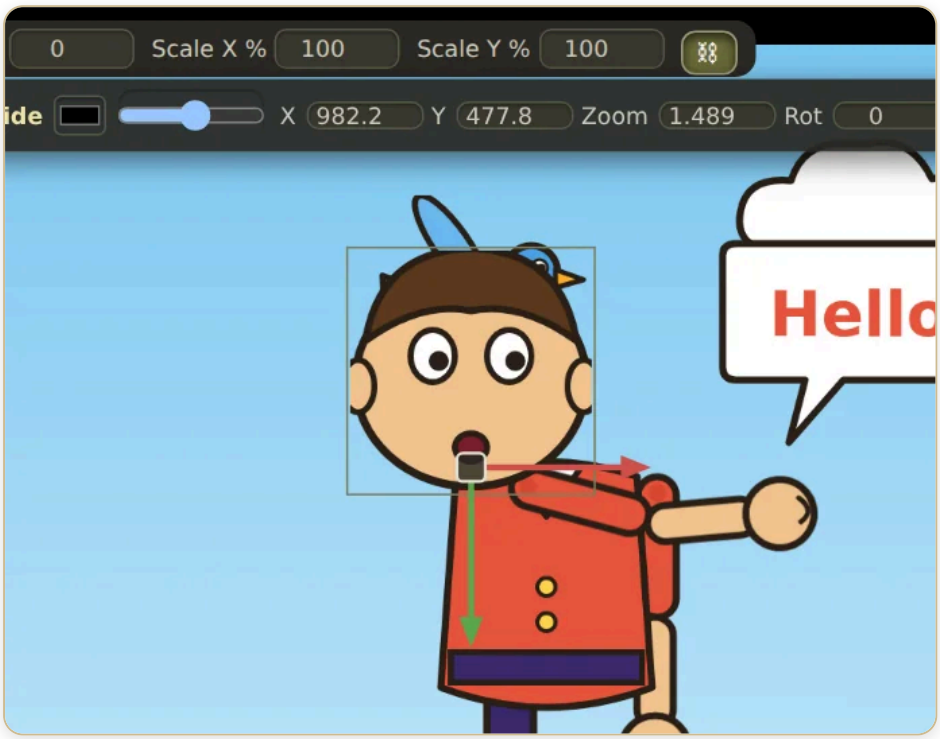
Characters as files

A finished rig is valuable. **Save Character** stores the folder, its layers and the rig; **File ▸ Import Character…** brings it into any other project. **Export Character Animation** also stores the keys, and **Load Character Animation…** loads both.

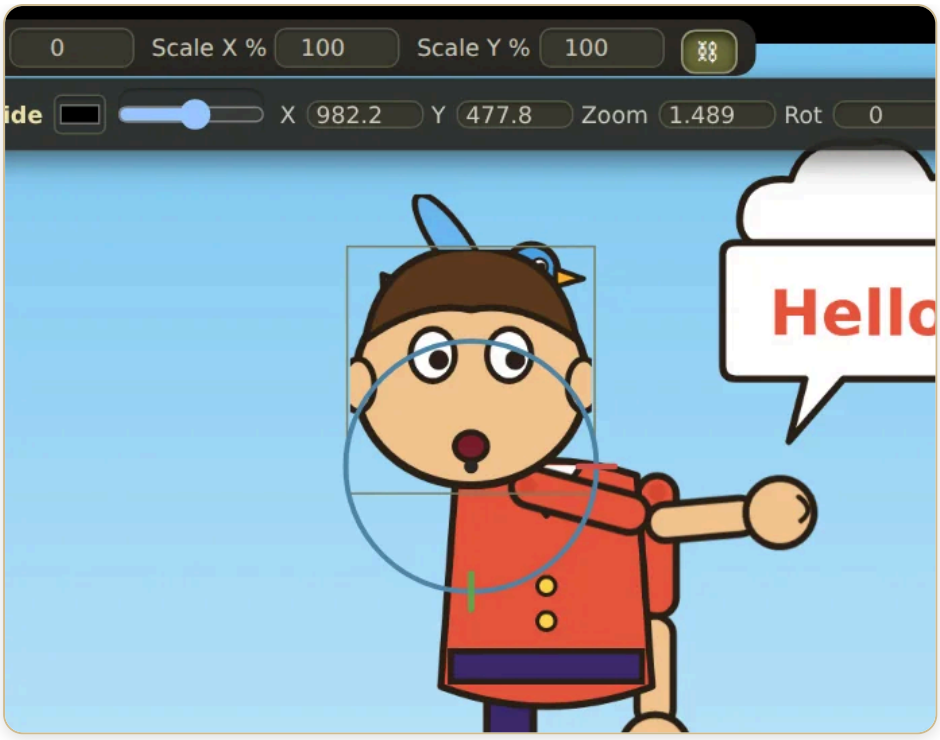
9 Transform, Pivot, Link and Offset

These four ideas turn a pile of pictures into a puppet.

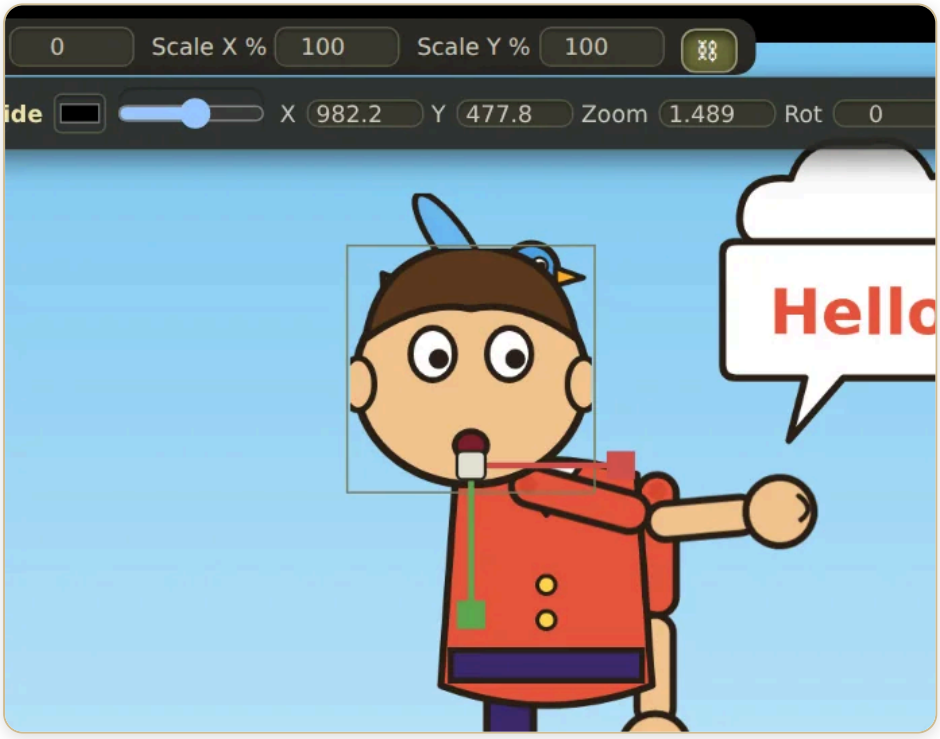
Move, Rotate, Scale



Move: red and green arrows, centre handle.



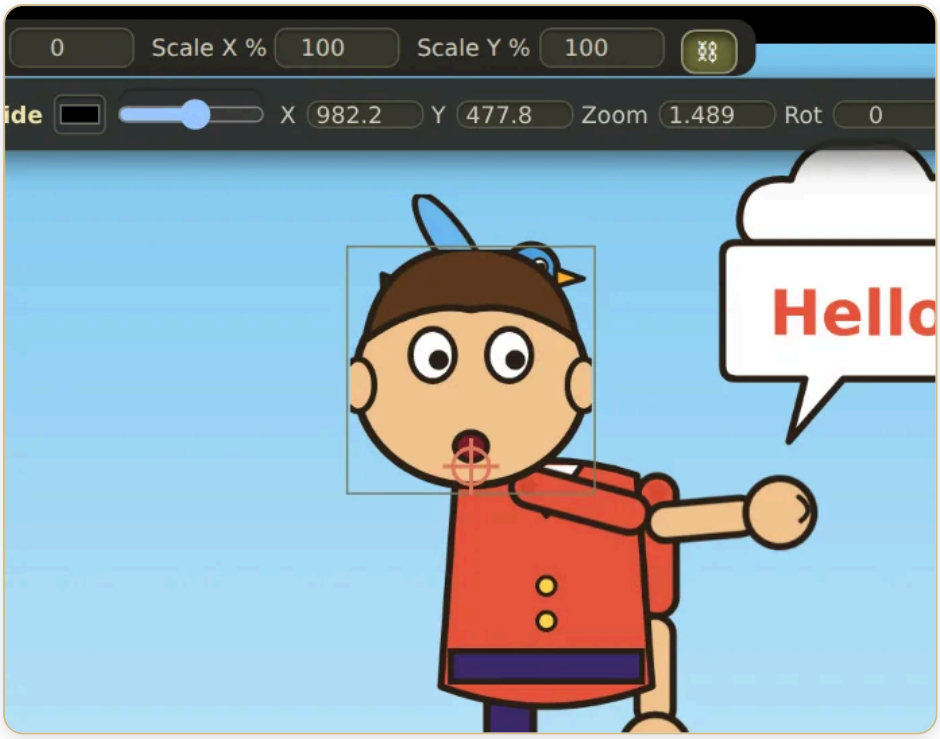
Rotate: the ring around the pivot. `Shift` *snaps to 15°.*



Scale: handles resize around the pivot.

Rotation and scaling always happen **around the pivot**. That is why the next tool matters so much.

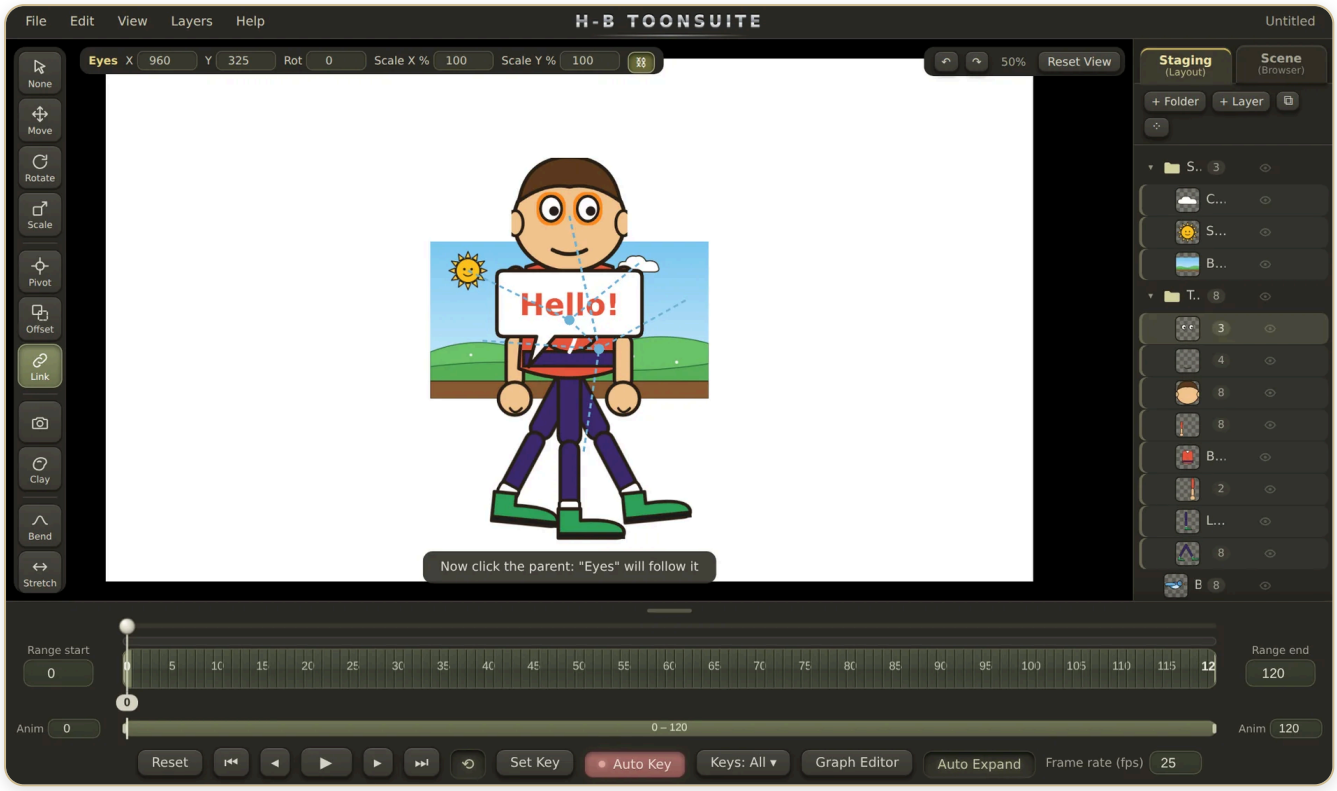
Pivot – the joint



Pivot tool: drag the little crosshair to the joint.

By default the pivot is the centre of the picture. A head tilting around its centre looks like a floating ball; tilting around the **neck** looks right. (Rotating a separate arm layer around its **shoulder** is the same idea.) Select the object, press P and drag the pivot to the joint (the neck, for a head). Changing a pivot is itself recorded in the **Pivot** key channel when Auto Key is on, so a pivot can even move during an animation.

Link – parent and child

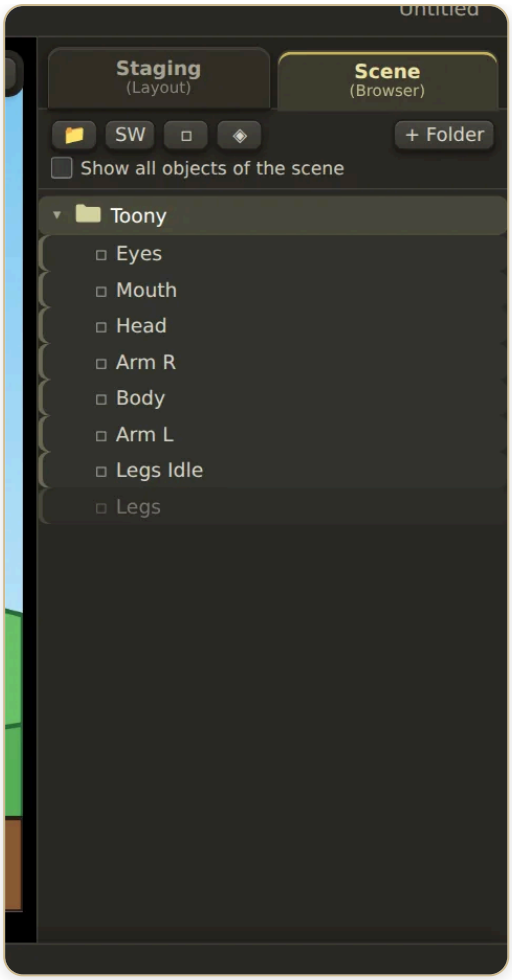


Link tool: the child (the Eyes layer) has been clicked and the program waits for the parent (the Head).

Linking two parts

1. Press L or click **Link**.
2. Click the **child** (for example the Eyes layer).
3. Click the **parent** (the Head). A message confirms the link.
4. Go back to **Move** W. Rotating the parent now carries the child along. Esc cancels a half-finished link.

When a folder is dropped, every layer is already linked to the master, so the whole character moves as one. You add extra links only where one layer must follow another: **Eyes and Mouth → Head** (so the face tilts and turns with the head), a hat → head, a tail → body. Hierarchies are shown in the **Scene** tab. The link itself is keyed in the **Link** channel, so a hand can grab a ball at frame 40 and let go at frame 60.



The Scene tab shows the hierarchy of what is on the stage.

> **Careful**
>
> A parent must not be linked to its own child. Plan the chain from the master outward: master → head → eyes and mouth.

Offset mode

Suppose you made a perfect walk cycle but want the character to lean a little in one scene. With **Auto Key off** and Offset 0 on, your changes are kept as an **offset on top of** the animation, with no keys touched. Reset Offset removes it; **Bake Offset into Keys** makes it permanent.

Reset, Freeze Transform and Import Animation

At the bottom of the side panel three buttons appear for the selected object:

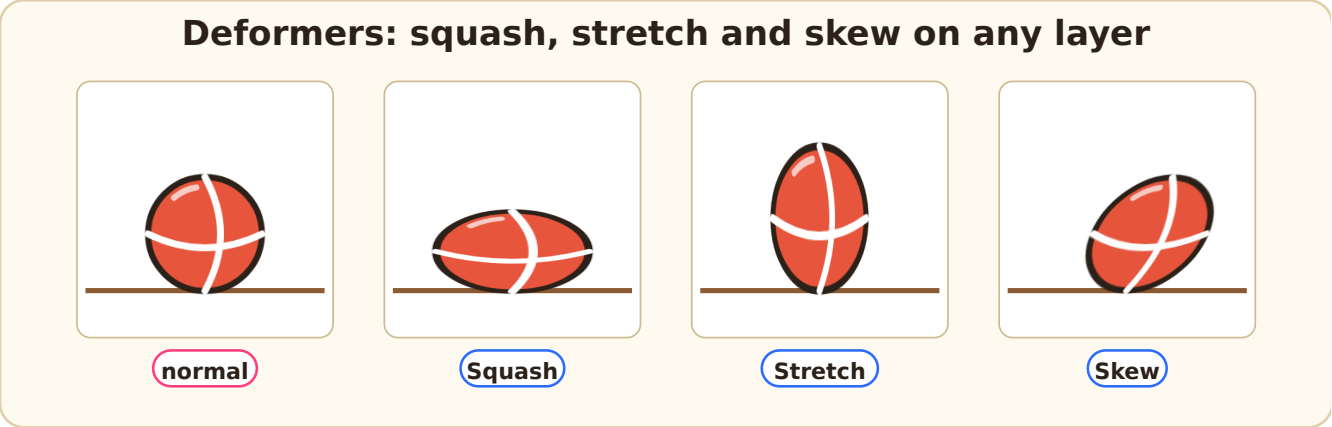
Button	Meaning
Reset	Return to the rest pose. On a folder only the master moves back; its layers keep their own local offsets.

Button	Meaning
Freeze Transform	Bake the offset, then make the current pose the new rest pose.
Import Animation…	Turn the images inside a layer into an animation (repeat / reverse) – handy for frame-by-frame sequences.

10 Deforming: Clay, Bend, Stretch, Squash, Skew, Hide, Flip

If the frames inside layers are the **actors**, the deformers are the **make-up and costume department**. They do two jobs that no drawing program can do as quickly:

1. **Matching layers** – making two separately drawn layers (a torso and a pair of legs, a head and a neck) meet perfectly so they look like one figure.
2. **Beautiful, expressive figures** – squash-and-stretch on impact, a gentle bend of a tail, a wobbly sun, a cheek that puffs.

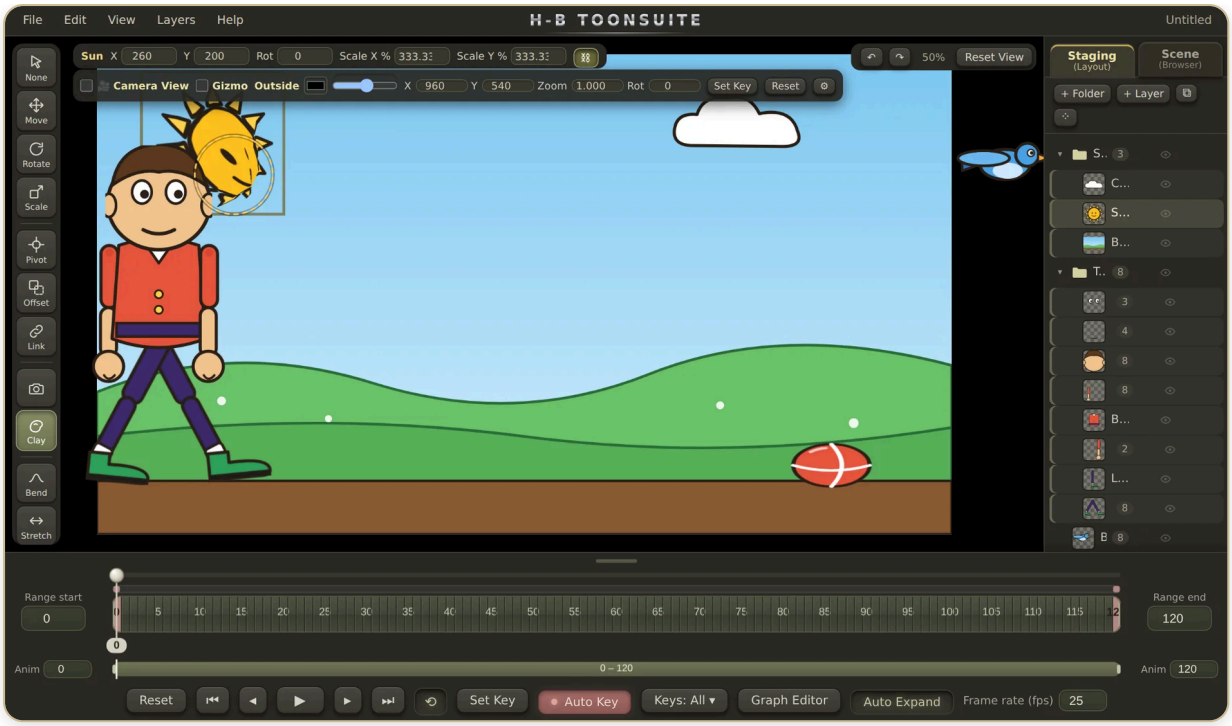


Squash, Stretch and Skew change a layer's shape without redrawing it.

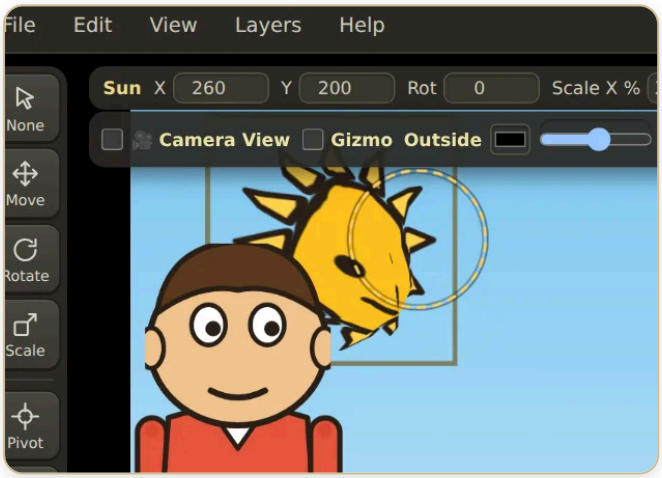
Clay – the sculpting brush

Clay behaves like the Liquify tool of a photo editor. The layer is a flexible pixel mesh, and a soft round brush pushes it around. Select the tool, then:

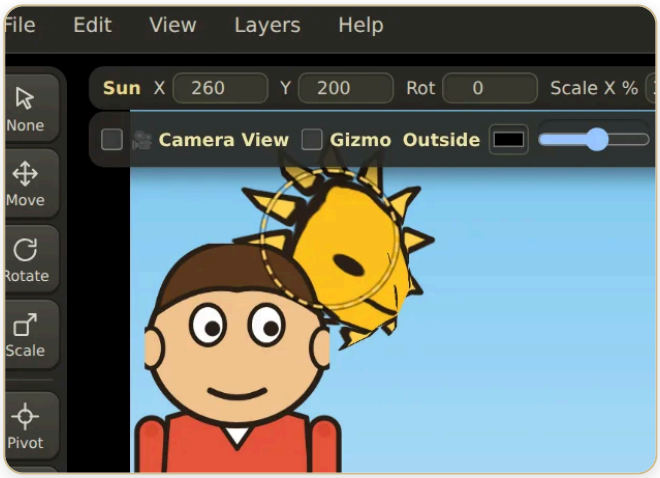
Gesture	Effect
Drag	**Push** the pixels along the stroke.
`Shift`+drag	**Twirl** – rotates the mesh under the brush; moving right turns it clockwise, left anticlockwise.
`Ctrl`+drag	Scales the mesh under the brush.
Wheel forward / back	**Pucker** (pull inward) / **Bloat** (push outward).
`Ctrl`+wheel	Brush size.
`Alt`+click	Choose the layer to sculpt when several overlap – it even reaches frozen or locked layers.



Clay in action: dragging over the sun pushes its shape like dough.

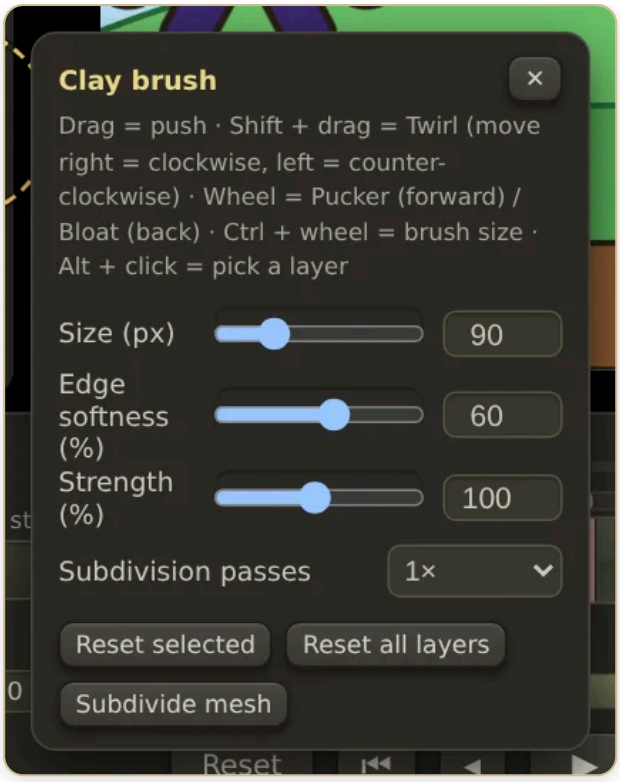


Twirl: `Shift`+drag.



Pucker / Bloat with the wheel.

Right-click the **Clay** button to open the **brush panel**:



The Clay brush panel.

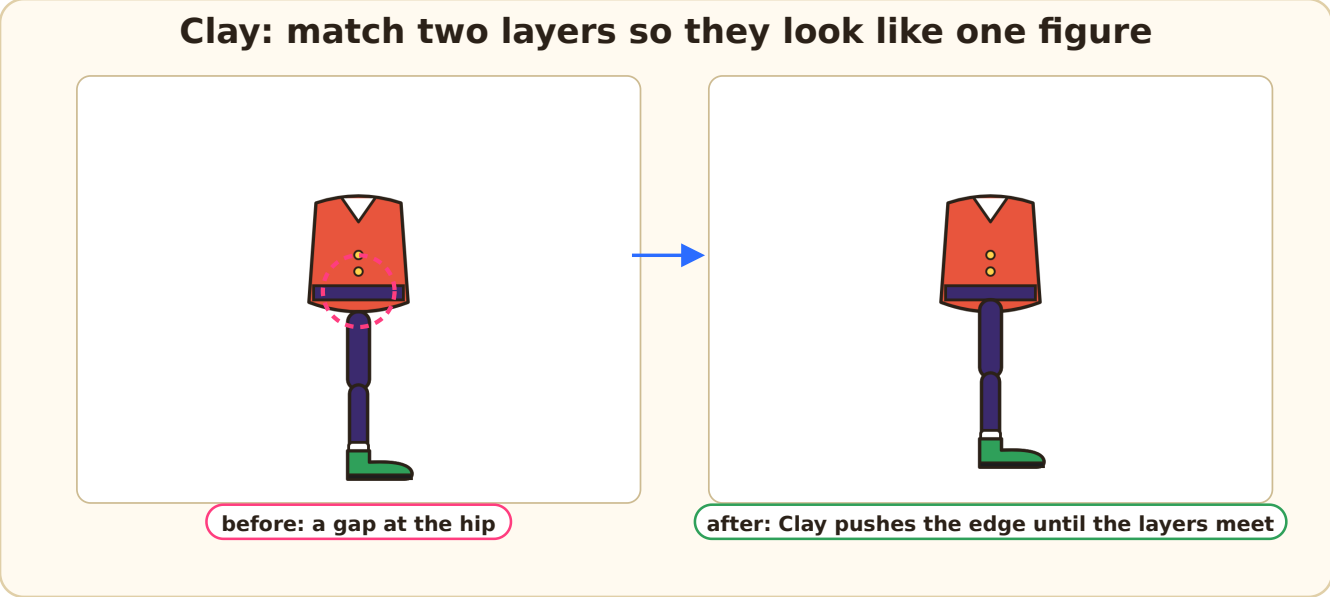
Setting	Meaning
Size (px)	Brush diameter.
Edge softness (%)	How gradually the effect fades toward the brush edge.
Strength (%)	How strongly a stroke moves the pixels.
Subdivision passes (1×–3×)	Smoothness of the mesh. Higher = finer, rounder results on big images.
Reset selected / Reset all layers	Remove the sculpting of the selected layer / of every layer.
Subdivide mesh	Make the mesh finer in one go, for detailed work.

> **Note**
>
> Each camera also has a **Clay / Bend look** setting (chapter 13): *High quality* keeps edges sharp at any zoom (slower on big images), *Preview* is lighter for fast sculpting.

Matching two layers with Clay

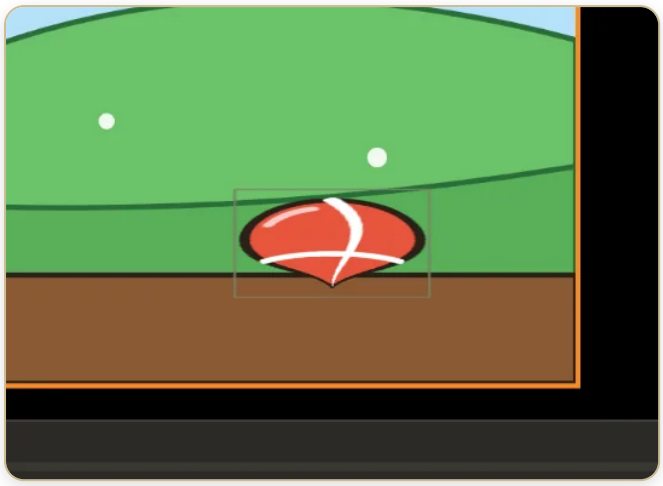
Say you drew the torso and the legs separately and a thin gap shows at the hip. Instead of redrawing, take a small Clay brush and **drag the edge of the legs up until it tucks under the belt**. Because the

sculpt belongs to the layer, it works for every frame of that layer's sequence if you do the same on each (and the mesh passes keep the line clean).

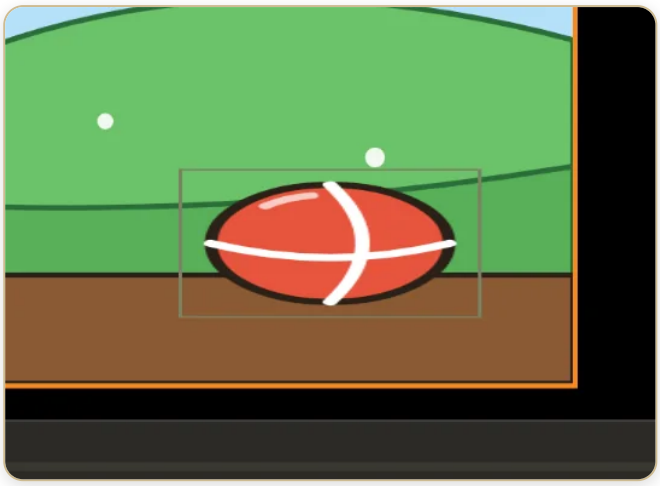


Clay closes the gap between the torso layer and the legs layer.

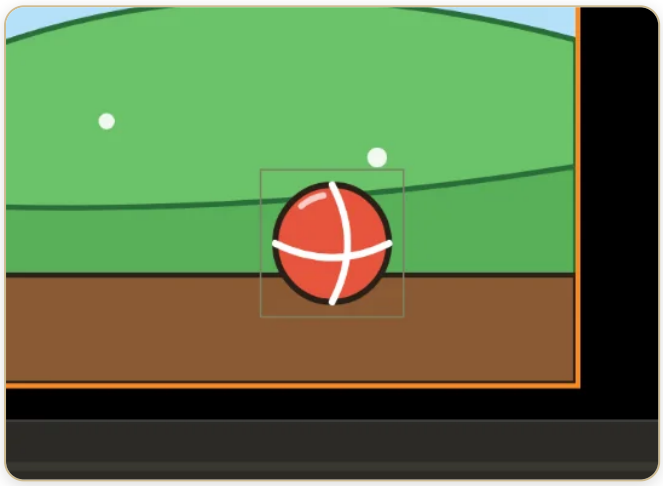
Bend, Stretch, Squash, Skew



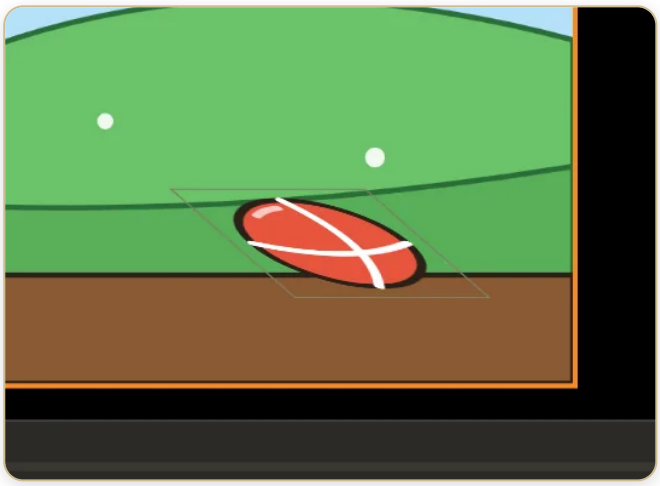
Bend – drag to curve the object.



Stretch – make it long and thin.



Squash – flatten it.



Skew – slant it.

Tool	Use it for
Bend B	Tails, hair, waving flags, a wobbling forearm.
Stretch T	A ball in the air, a jump, a quick pull.
Squash U	Impacts: landing, bouncing balls, a blink-flatten.
Skew K	Slanting a building, a speed-lean, a swaying tree.

With Auto Key on, all four are recorded in the **Deform** channel, so you can squash on frame 30 and return to normal on frame 34. The pivot decides where the shape “sits” – keep the pivot at the ground contact so a squashed ball stays on the floor.

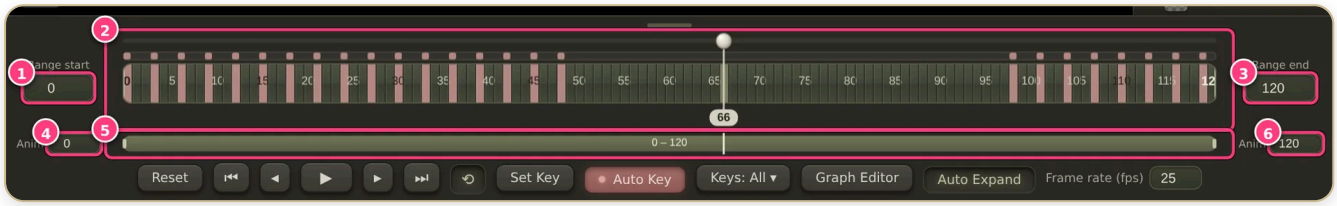
> **Note**
>
> Clay sculpting is saved with the object as its shape. Plan animated wobble with Bend and Skew keys; use Clay for the base shape and for matching.

Hide and Flip

- `Hide` shows or hides the selected layer or folder and is stored in the **Hide** channel – a ball can vanish at frame 108, and the face layers can hide while the head spins.
- `Flip H` mirrors horizontally (stored in the **Flip** channel). Flipping the master folder turns the whole character around – the cheapest way to make him walk back.
- `Flip V` mirrors vertically – useful for reflections.

11 Keys and the Timeline

Animation means **changing a value between two moments**. A **key** records the value at one frame; the program fills the frames between keys (“in-betweening”).

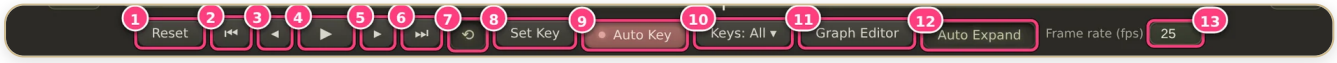


The timeline area. Selecting a layer shows its keys – the stepped image keys of an imported cycle appear as a dense row.

1. **Range start** – first frame of the **working range** (what you see and play).
2. The **frame strip**: pink marks are keys, the white line is the playhead.
3. **Range end** – last frame of the working range.
4. **Anim** start – first frame of the whole animation.
5. The **animation range bar**; drag its ends to cut the animation.
6. **Anim** end – last frame of the whole animation.

Auto Key and Set Key

- **Auto Key ON** (the red dot is lit): every change you make is keyed at the current frame in its own channel. This is the normal way to work.
- **Auto Key OFF**: changes are not keyed. Press **Set Key** `S` to key manually; on animated objects unkeyed changes become an offset.

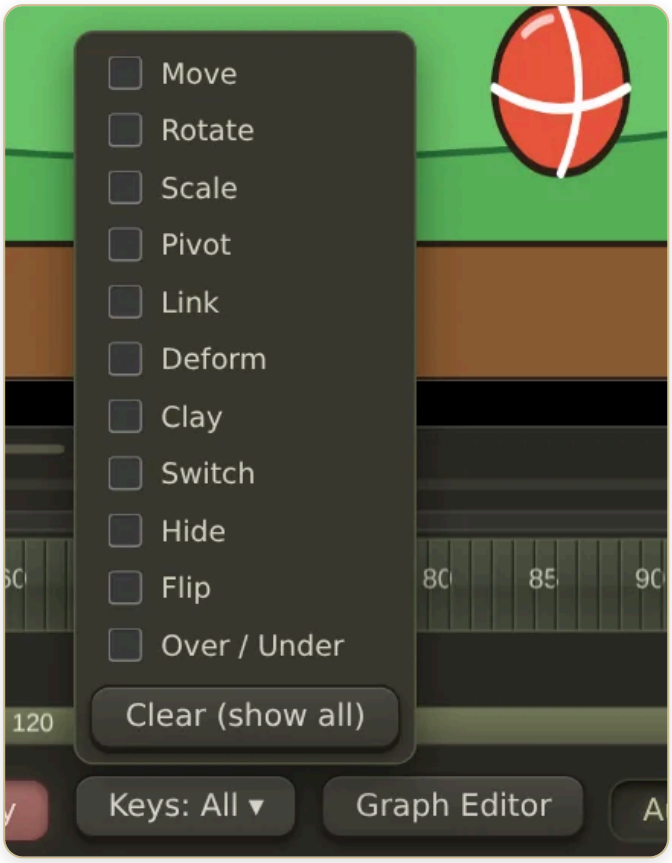


Transport and key controls.

1. **Reset** – back to the first frame.
2. First frame `Home`.
3. Previous frame `Left`.
4. Play / Pause `Space`.
5. Next frame `Right`.
6. Last frame `End`.
7. **Loop** playback on/off.
8. **Set Key**.
9. **Auto Key**.
10. **Keys: All ▾** – key filter.
11. **Graph Editor** `G`.
12. **Auto Expand** – the animation range grows when you key beyond its end.
13. **Frame rate (fps)**.

Key channels

Every kind of change is stored in its own channel, so a key for rotation does not disturb a key for position. The channels are: **Move, Rotate, Scale, Deform, Pivot, Link, Flip, Hide, Switch, Over / Under**. Images, switch folders, hide, flip and layer order are **stepped** (they jump rather than glide).



The key filter menu next to "Keys: All".

Use the filter to show only one kind of key – for example only **Rotate** – so the timeline is not cluttered. **Clear (show all)** returns to everything.

Working with keys

Action	How
Set a key on a frame	Double-click the frame on the strip, or **Set Key** `S`
Select keys	Click; `Shift+click` toggles; drag a box on the strip; `Ctrl+A` selects all
Move keys	Drag the **yellow handle** that appears over the selected keys
Copy / cut / paste	`Ctrl+C / X / V` with the mouse over the timeline; paste goes to the playhead
Reverse a motion	Select keys, `Shift+R`

Action	How
Delete keys	`Del`
Key menu (cut/copy/paste/tangents)	Right-click a key
Zoom / pan the timeline	Wheel / middle-drag

Frames on the timeline

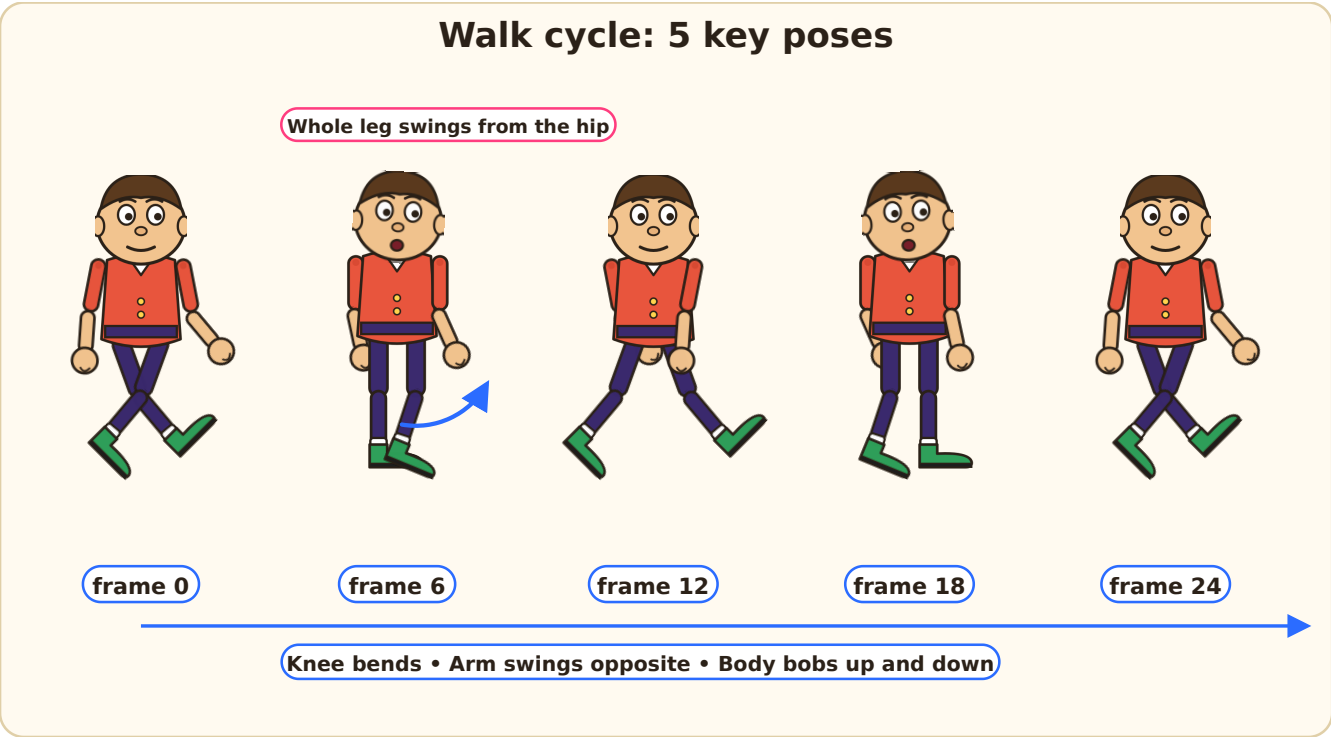
Every image change of a layer is a **stepped key** in the **Switch/Images** part of its keys: the value does not glide, it jumps. Import Animation fills the timeline with those keys in one go; you can then select, move, copy or delete them like any other key. A big **frame number** flashes while you scrub, and the **2 / 8**-style counter tells you which image of a layer is showing.

Making a cycle

A walk is the same few drawings again and again. There are two ways to build the repetition:

1. **Import Animation** (best): load the frames into the layer, set **Repeat** and **Frames per image**, press Import. Done.
2. **Copy and paste keys**: key one cycle (for example frames 0–23), select those keys, copy them, move the playhead to frame 24 and paste. Repeat as often as needed.

The first and last drawing of a cycle should flow into each other so the loop is seamless.



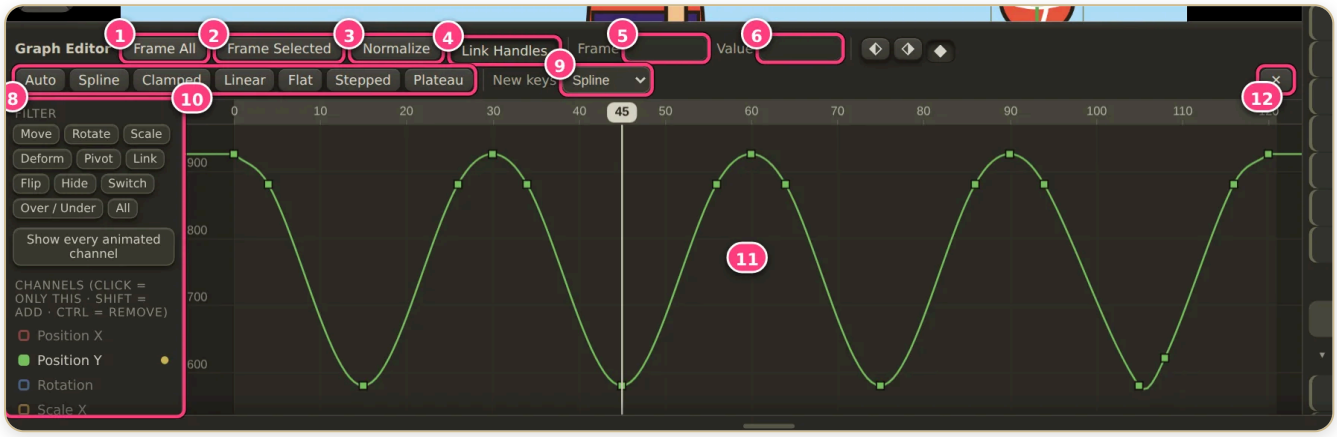
A walk cycle that lives inside ONE layer: eight drawings of the legs, each held for three frames.

Tip

Walk **in place** (the frames inside the legs layer) and move the master folder across the stage with two keys (start and end, linear). It is faster and easier to control than anything else.

12 The Graph Editor

The timeline tells you **when** a key happens; the Graph Editor shows **how** the value travels between keys. A steep curve is fast, a flat curve is slow. Open it with Graph Editor or G.



The Graph Editor header and channel list.

1. Frame All A – fit every curve in view.
2. Frame Selected F – zoom to the selected keys.
3. Normalize – scale curves to the same height so small and large values can be compared.
4. Link Handles – move both tangent handles together.
5. **Frame** of the selected key (type to move it).
6. **Value** of the selected key.
7. Buttons that control how the handles of the selected key behave.
8. Tangent types for the selected keys.
9. **New keys** – the default type for keys you create from now on.
10. Channel filter and channel list.
11. The curve view with the playhead.
12. Close the editor.

The seven tangent types

Type	Behaviour	Good for
Auto	The program chooses a smooth slope.	General use
Spline	Smooth curve through all keys (the default).	Organic motion, walk cycles
Clamped	Smooth, but never overshoots the key values.	Stops without wobble
Linear	Straight line, constant speed.	Camera pans, travel across the stage
Flat	Horizontal handles – ease in and ease out.	Holds, anticipation

Type	Behaviour	Good for
Stepped	Value jumps at the next key.	Pops, flicker, “held” poses
Plateau	Smooth but flat at every key, so the curve “rests” on each.	Held pose then move

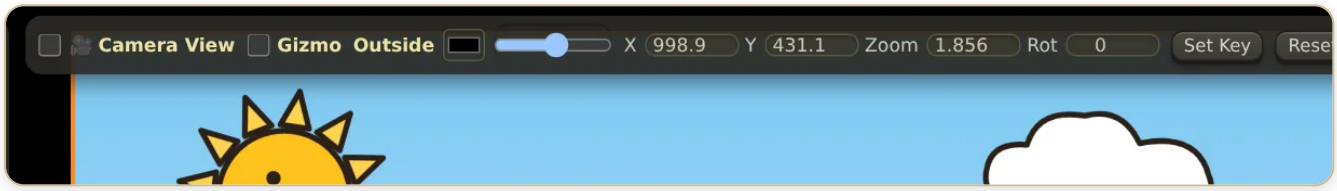
Action	How
Show a curve	Click its channel name; `Shift` adds; `Ctrl` removes
Zoom	Wheel (time + value), `Shift+Wheel` value only, `Ctrl+Wheel` time only
Pan	Middle-drag
Select all graph keys	`Ctrl+A`

> **Tip**
>
> A bouncing ball: use **Flat/Plateau** at the top of the arc (slow, floaty) and **Linear** or **Stepped** at the ground contact (sharp). The graph will show the classic “sawtooth with round tops”.

13 The Camera and Effects

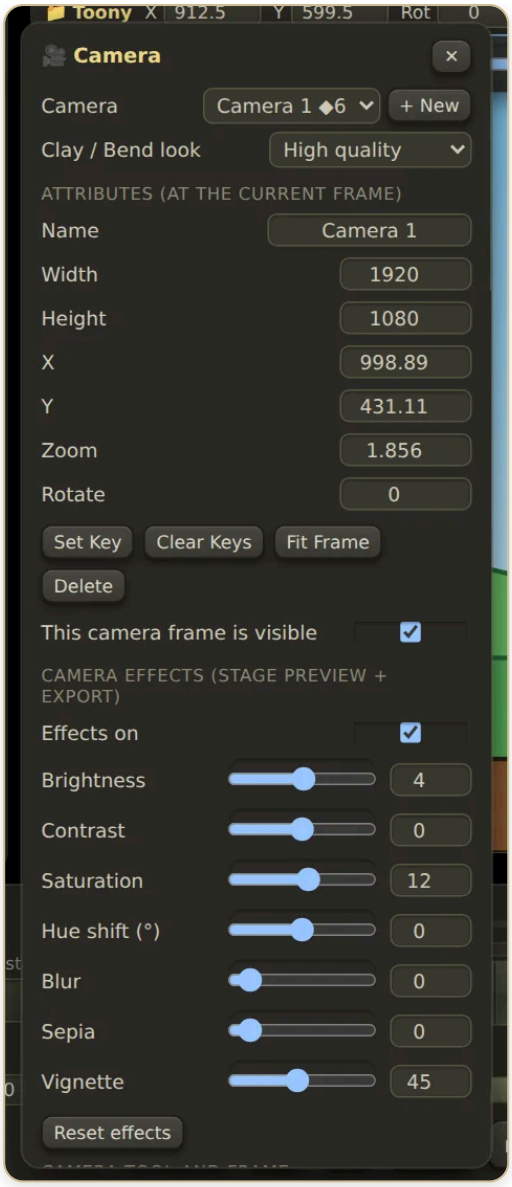
Without a camera the exported video is the whole stage frame. A camera lets you **zoom into a face, pan with the character or tilt the world** – and every move is animatable.

- Press `C` or click the camera icon to switch the **camera tool** on. Then **drag** to move the camera, `Shift` **+ drag** to rotate it (with `Ctrl` in 15° steps) and use the **wheel** to zoom.
- **Right-click** the camera icon (or use `Layers ▸ Camera Settings…`) to open the camera panel.
- Create more cameras with `+ New` or `Layers ▸ New Camera` and switch between them in the panel.



The camera bar above the stage: Camera View, Gizmo, the Outside colour and opacity, and the current camera values.

Control	Meaning
Camera View	Look through the camera: you see exactly what will be exported.
Gizmo	Show the camera schematic and its handles on the stage so you can grab and move the camera directly.
Outside (colour + opacity)	Colour and strength of the dark veil over the area outside the camera frame.



The camera panel: attributes at the current frame, keys and effects.

Control	Meaning
Name, Width, Height	Camera name and output size in pixels.
Clay / Bend look	*High quality*: sharp edges that follow the stage zoom (slower on big images). *Preview*: lighter, for fast sculpting.
X, Y, Zoom, Rotate	Position of the camera centre, magnification and tilt **at the current frame**.
Set Key / Clear Keys	Key the camera now / remove all camera keys.
Fit Frame	Make the camera cover exactly the stage frame.
Delete	Remove this camera.
This camera frame is visible	Show its rectangle on the stage.
Effects on	Master switch for the effects below – they show on the stage and in the export.

Control	Meaning
Brightness, Contrast, Saturation	Basic colour correction.
Hue shift (°), Sepia	Rotate all colours / old-photo tone.
Blur, Vignette	Soften the picture / darken the edges to focus on the centre.
Reset effects	Back to neutral.

In Project Settings you can also set the **new camera size**, whether the selected camera frame is shown, how much to **darken the area outside the camera frame** (opacity %) and how long the camera name label stays on the stage.

> **Tip**
> A slow zoom-in during a line of dialogue plus a gentle **Vignette** is the cheapest “drama” you can add to a limited-animation scene.

14 Reusing Work: Clips and Characters

Limited animation is about **not redrawing**. H-B ToonSuite gives you four levels of reuse.

Level	What is reused	How
Frames in a layer	A drawn cycle stored inside one layer.	**Import Animation**: Repeat, Frames per image, Load reversed, Start at frame (chapter 7).
Keys	A few frames of motion inside one project.	Copy / paste keys on the timeline (chapter 11).
Clip	The animation of a selected set of objects, as a file.	Right-click an object ▸ **Export Clip (animation of this set)…**, later ▸ **Import Clip at this frame…**.
Character	The whole puppet: folder, layers, images and rig – with or without animation.	**Save Character…** / **Import Character…**; **Export Character Animation…** / **Load Character Animation…**.

Build a “walk clip” library

1. Key one clean walk cycle on a character (for example 24 frames).
2. Select the character, right-click ▸ **Export Clip…** and save “walk.clip”.
3. In any scene, move the playhead to the frame where the walk should start, select the character and use **Import Clip at this frame…**.
4. Repeat for “wave.clip”, “talk.clip”, “jump.clip”. Soon a new episode is mostly assembly.

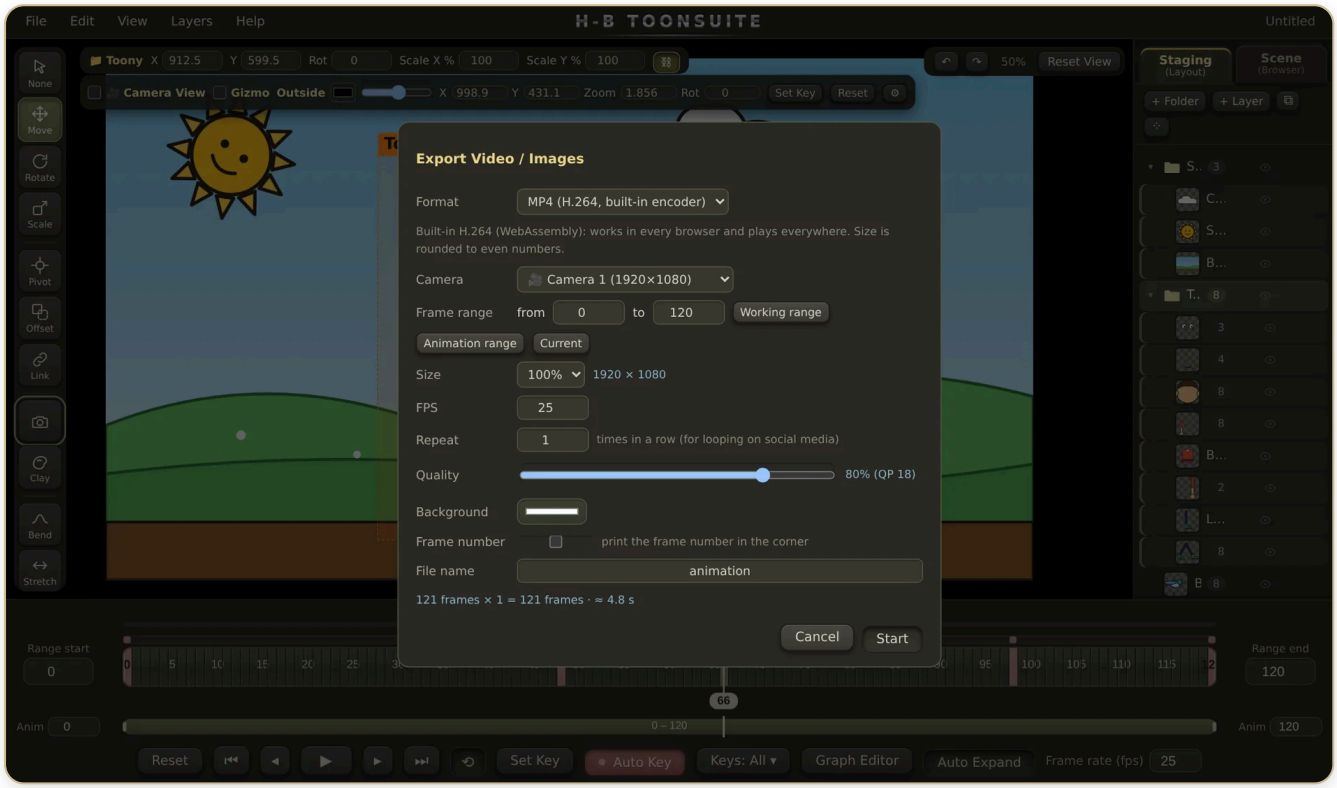
Note

A clip applies to a character with the **same structure** (same part names and hierarchy). That is another reason to keep names tidy and identical across characters.

15 Export, Settings and Performance

Exporting a video or images

Choose File ▸ Export Video / Images… Ctrl+E.



The export dialog.

Setting	Meaning
Format	**MP4** (H.264, built-in encoder – plays everywhere), **WebM** (VP9), **JPG sequence**, **PNG sequence** (one picture per frame).
Camera	Export the full stage frame or one of your cameras.
Frame range from / to	Buttons Animation range, Current and Working range fill the numbers for you.
Size	100 %, 75 %, 50 % or 25 % of the frame. Use 50 % for quick tests.
FPS	Frames per second of the exported file.
Repeat	Repeats the animation N times in a row – good for short loops on social media.
Quality	Compression quality of video or JPG.
Background	Colour behind transparent areas.

Setting	Meaning
Frame number	Print the frame number in a corner – useful for review notes.
File name	Name of the result.
Summary line	For example “101 frames × 1 = 101 frames ≈ 4.0 s”.

> **Note**
> H.264 needs even width and height; the program rounds the size for you. Export runs in the browser, so keep the tab open and visible until it finishes.

Project Settings

Settings ▸ Project tab.

Setting	Meaning
Project name	Shown at the top right of the window.
Frame size (px)	Size of the stage frame – default 1920 × 1080.
Show the intro screen…	Welcome screen on/off at start-up.
Save Plus (numbered versions)	Each save creates a new numbered file.
Frame rate (fps)	Playback speed – default 25.
Range start / end	First and last **working** frames.

Setting	Meaning
Auto Key, Loop playback, Auto Expand animation range	Defaults for the three switches at the bottom of the screen.
Default key interpolation	The tangent type new keys start with (Spline by default).
Proxy images Open…	Opens the proxy dialog (below).
Cameras section	New camera size, show camera frame, darkness outside the frame, label time.

Colours, shortcuts and app zoom

Settings ▸ Colors tab.

The **Colors** tab lets you recolour the interface: background, panels, buttons, text, accent, stage frame, timeline keys, playhead, camera area and more. **Reset to defaults** restores the original theme. The **App** tab holds **App zoom (%)** and the buttons to **Export… / Import…** a settings file (colours, zoom and shortcuts) so you can move your setup to another computer.

Settings ▸ Shortcuts tab. Click any shortcut to change it.

In the **Shortcuts** tab click a key box and press the new combination. ↺ resets one shortcut, ⊘ removes it. A search box finds actions by name. Mouse gestures can be changed too. The full default list is in the appendix.

Proxy images – faster playback

Big images make playback heavy. **Proxy images** shows scaled-down copies on the stage (you choose the **longest side in pixels**), while your originals stay untouched and are always saved and exported. Toggle it with `View ▸ Proxy Images` and set its size in `View ▸ Proxy Settings…`. The **Clean** button clears the cached copies.

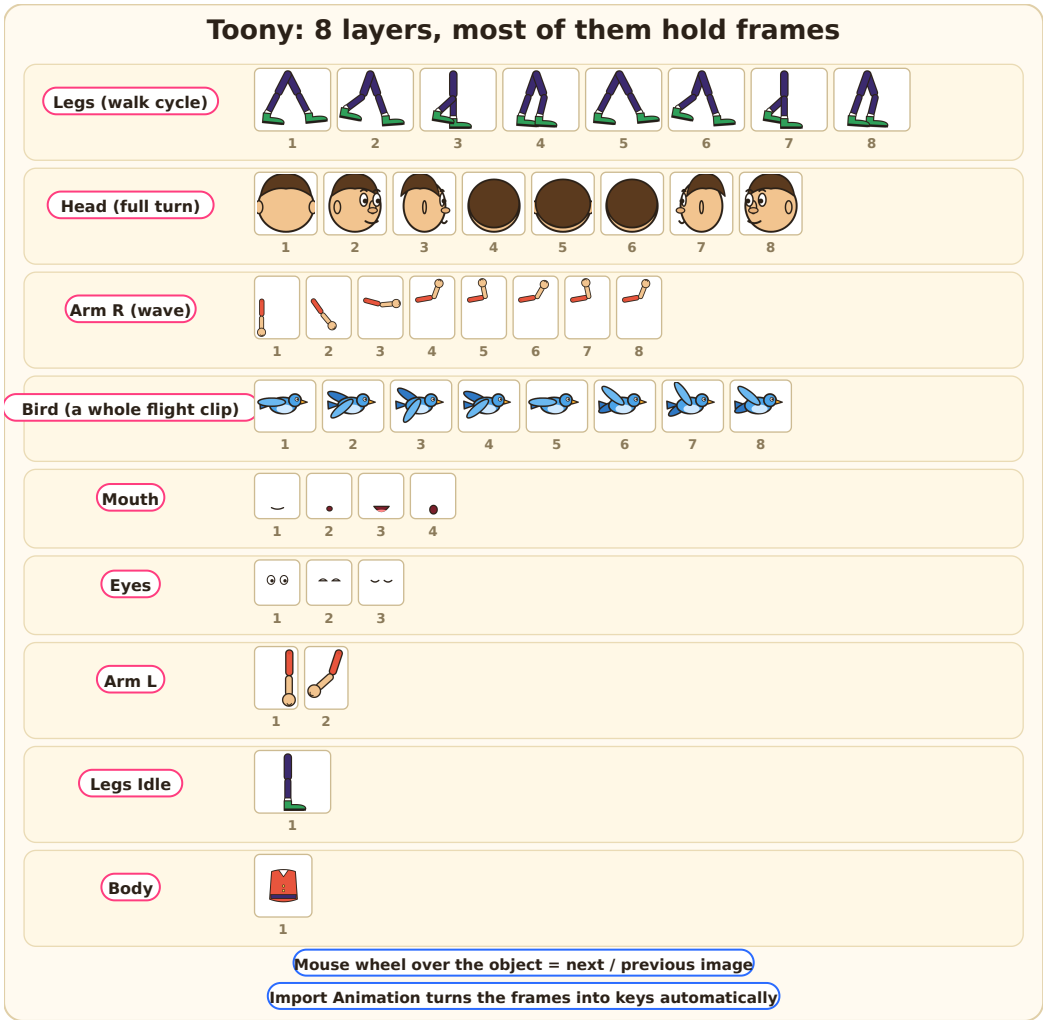
> **Tip**
>
> If playback stutters, turn Proxy Images on and lower the stage zoom. Export is not affected.

16 Anatomy of a Frame-Based Character

Meet **Toony**, the original character used in this book, and **Chirp**, a little bird. They are built the way a limited-animation figure should be: a few layers, **most of which carry their own frames**. This chapter explains them with arrows and signs. Read the diagrams like a map:

Sign	Meaning
Numbered white card	one frame (image) inside a layer, in playing order
Pink label	name of a layer or a rule
Blue label / arrow	time direction or an explanation of timing
Pink arrow	“follows” (link) or “repeat”

1. Eight layers, most with frames



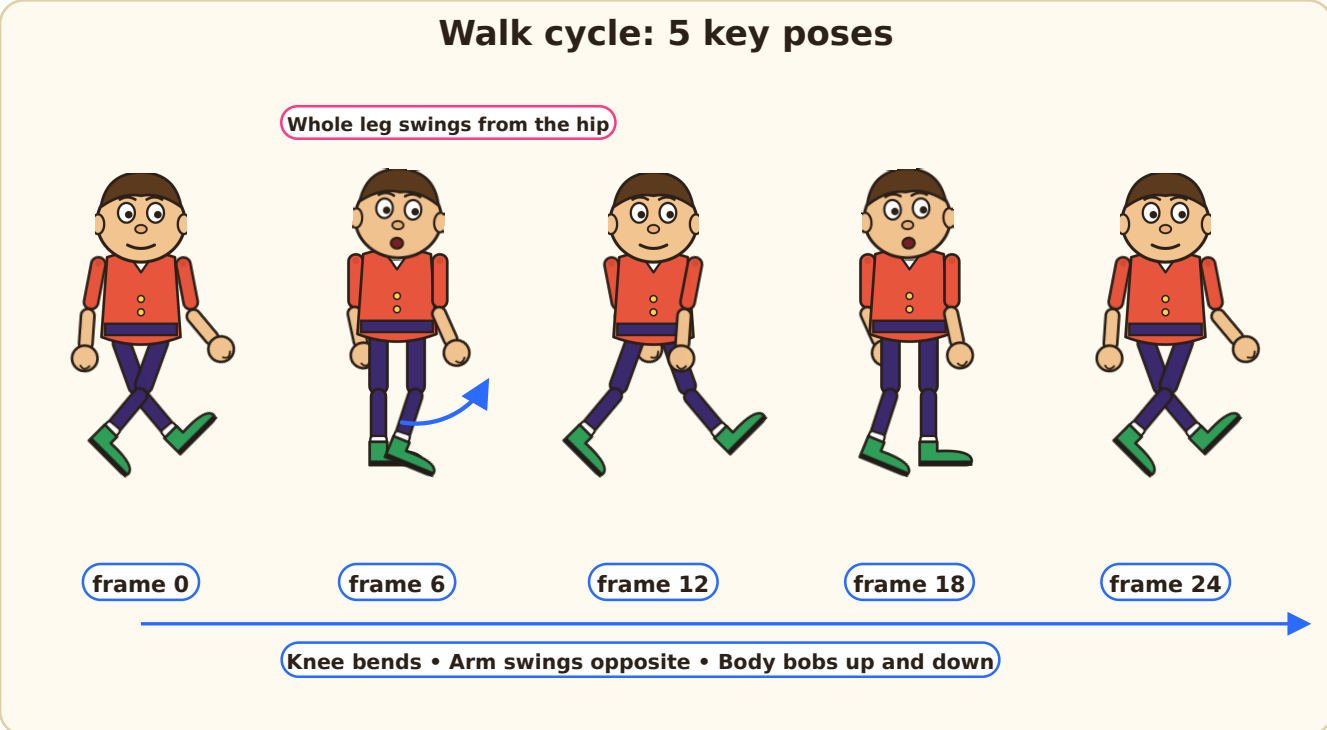
Every row is one layer; every card inside a row is one frame of that layer.

Look at how the work is divided. The **Body** never changes: one layer, one image. The **Legs** carry the whole walk, the **Head** carries a whole turn, the **Arm R** carries a whole wave and the **Bird** is a complete flight loop. **Eyes** and **Mouth** are small families of expressions, and **Legs Idle** is the standing pose. Eight layers, but dozens of drawings – and each one drawn only once.

2. How to split your own character

- Look for the natural **seams** of the design: the collar, the belt, the shoulder. Cut there.
- **Same canvas, same registration.** Export every frame of a layer on the same canvas size with the part in the same place. Then frames never jitter when they change.
- Keep the **biggest static piece (the torso)** as the biggest layer; it is a good anchor.
- Draw only what you need: a head that only talks needs mouth frames, not a whole new head.

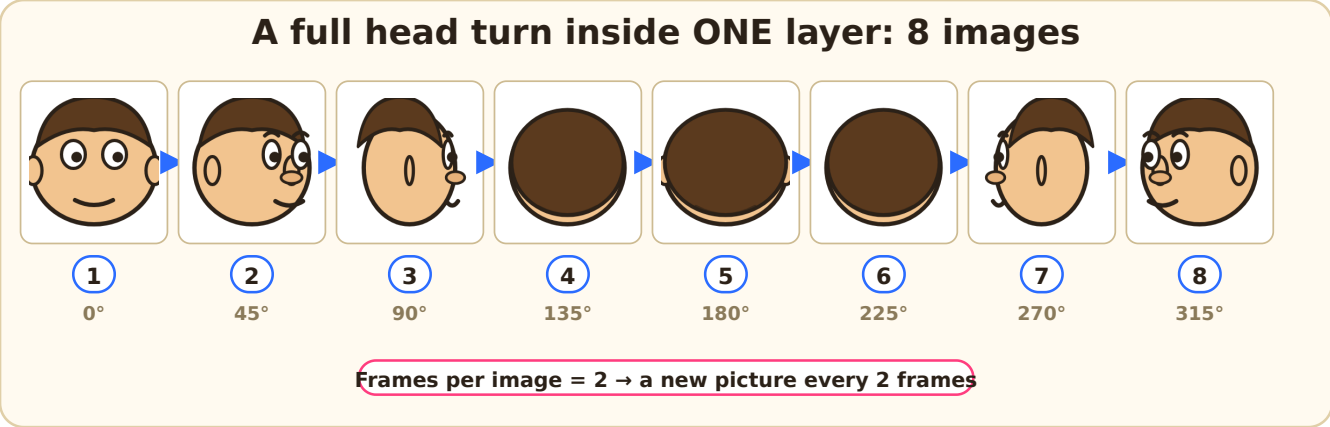
3. Walking – the legs layer



Eight drawings of the legs inside ONE layer. Import Animation holds each for 3 frames and repeats the loop.

Setting	Value	Result
Images in the layer	8	one complete step of both legs
Frames per image	3	24 frames per cycle (about 1 second)
Repeat	2	48 frames of walking
Master folder	moves 800 px	the character crosses the stage

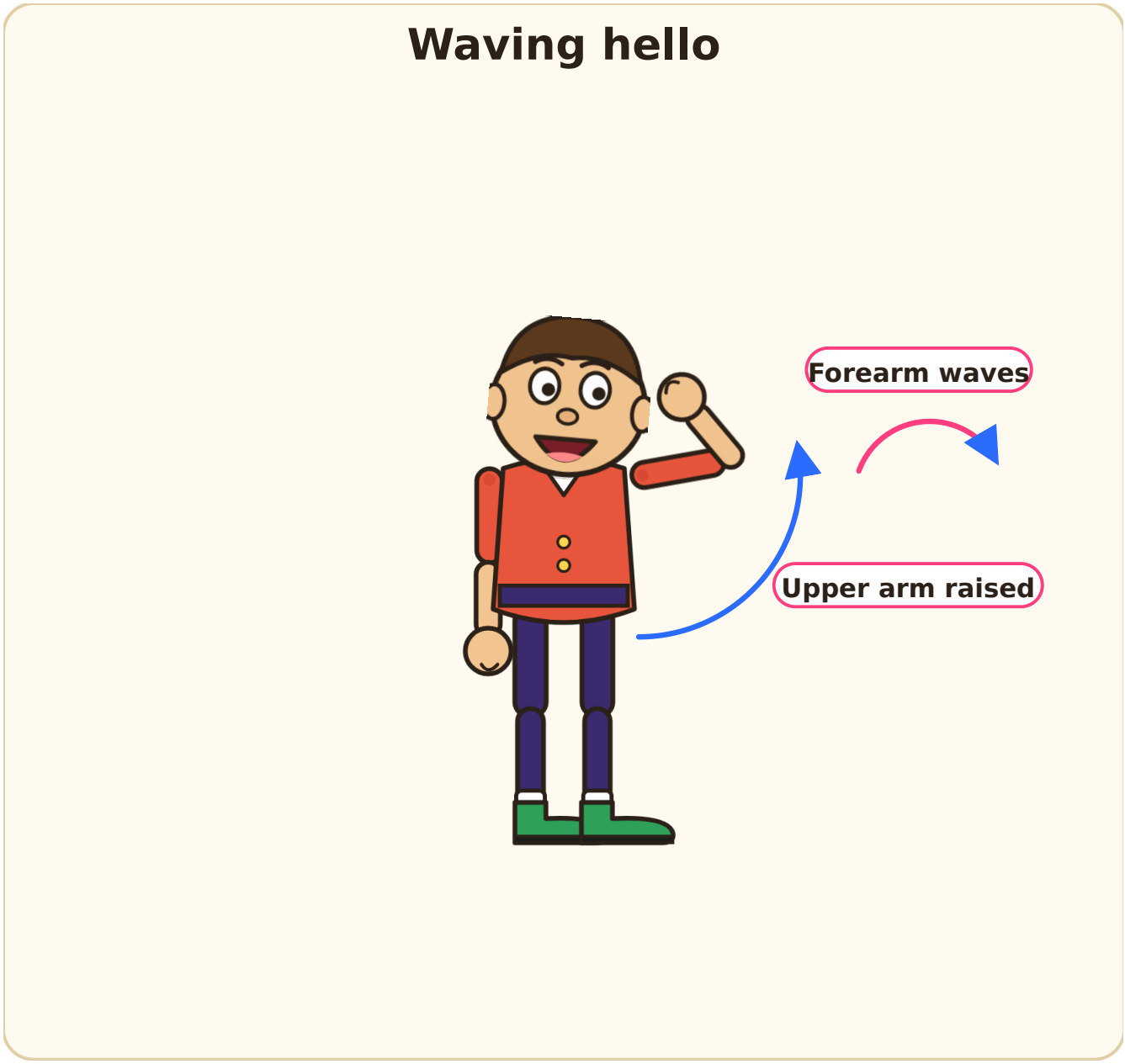
4. Turning the head – a full turn in one layer



Eight views of the head: front, three-quarter, profile, back and round again.

Frame 1 is the plain front head (without face) so the separate **Eyes** and **Mouth** layers can sit on it and talk. The other seven frames have their face baked in, so while the head turns we simply **hide** Eyes and Mouth with the Hide tool and show them again when the head is back to the front.

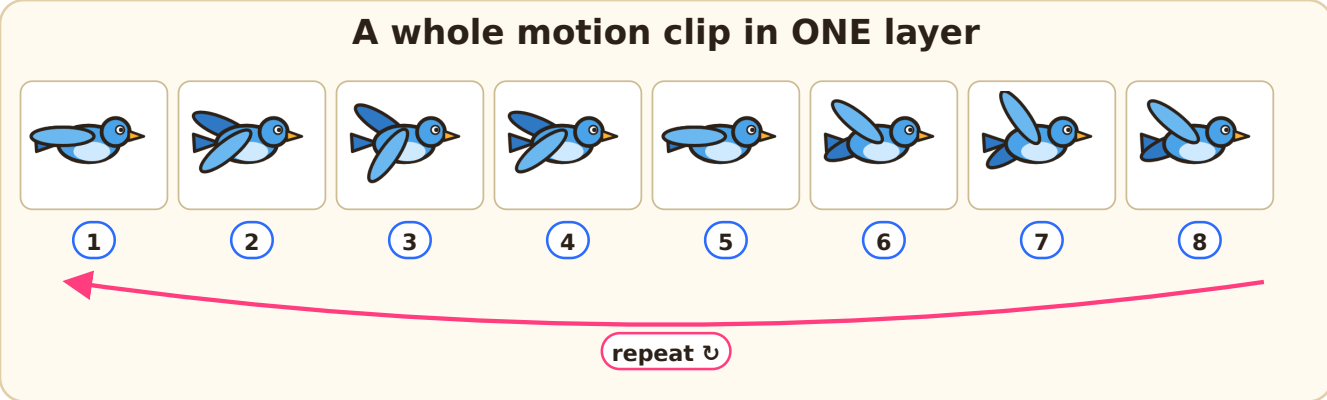
5. A gesture – waving hello



The arm layer holds the whole wave: raise, then swing the forearm.

Import the eight arm frames once at the start of the greeting. When the greeting is over, import them again with **Load reversed** to lower the arm smoothly – no new drawings needed.

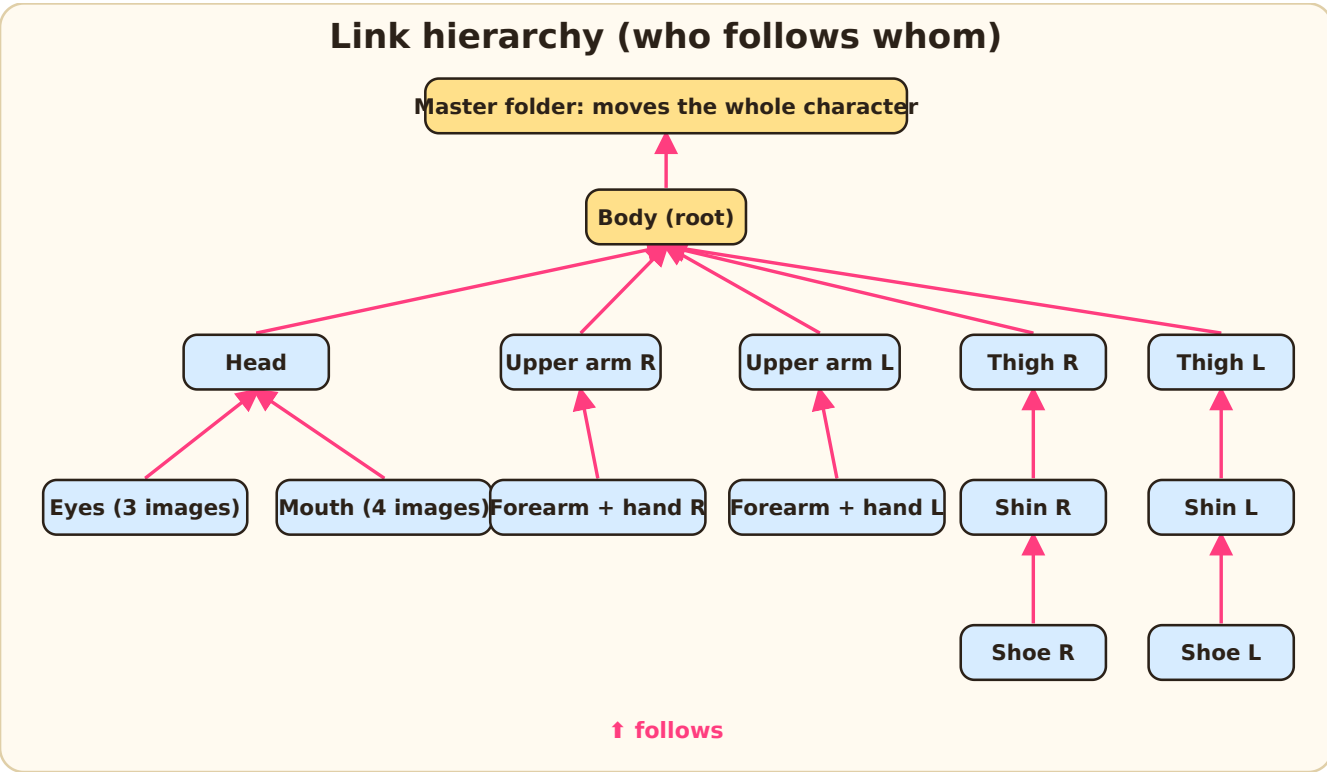
6. A whole clip in one layer



Chirp the bird: eight flap drawings in one layer, repeated in a loop.

The bird is the purest example. One layer, eight drawings. Import Animation with **Repeat 7** and **Frames per image 2** gives 112 frames of flapping; two keys on **Position X** carry the bird across the sky. The same trick works for a running dog, a spinning wheel, a rippling flag – anything that can be drawn as a loop.

7. Who follows whom, and the finishing touch

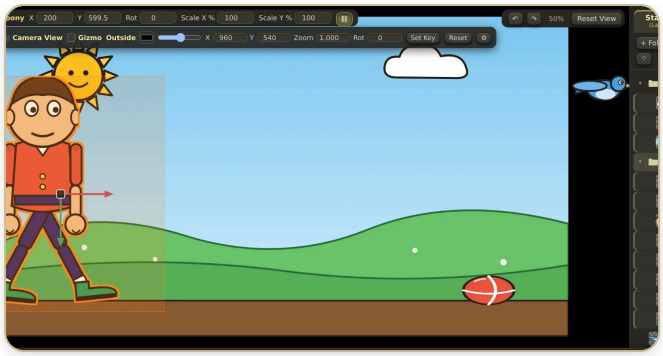


The link tree. The master carries everything; Eyes and Mouth also follow the Head.

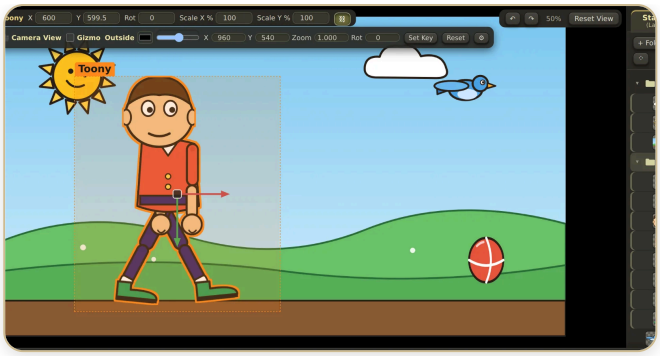
Finally the deformers: use **Clay** to make the legs meet the torso without a gap, and **Squash / Stretch** on the ball for life (see chapter 10).

17 Complete Project: “Toony Says Hello”

Now we build a real 4.8-second cartoon from zero to a finished video. **Toony walks in, stops, waves and says “Hello!”, then says “Bye!”, spins his head, turns around and walks away** while a bird flies across the sky, a ball bounces, the sun spins and the camera pushes in on his face. Nearly every feature of the program is used at least once – and almost all the animation comes from **frames stored inside layers**.

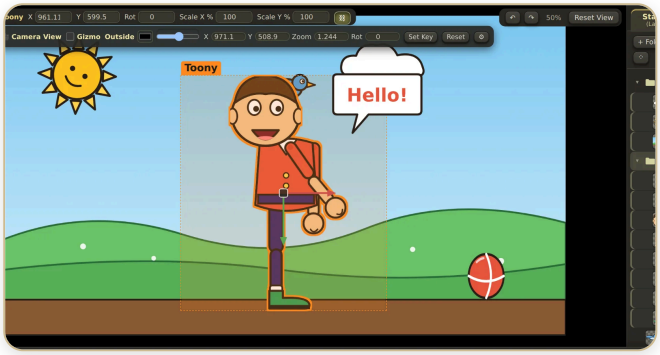


Frame 0 – walking in, ball squashed.

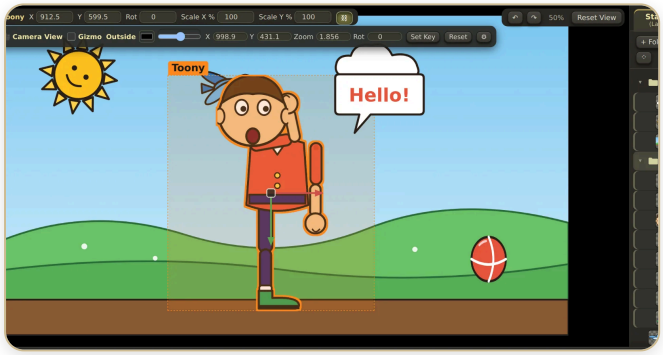


Frame 24 – mid-walk, bird flying.

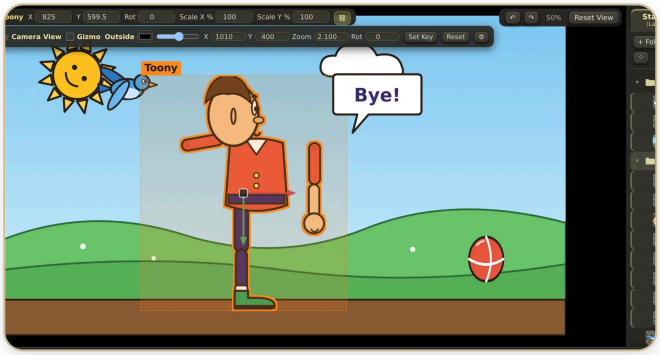
Frame 48 – stops: the idle legs take over.



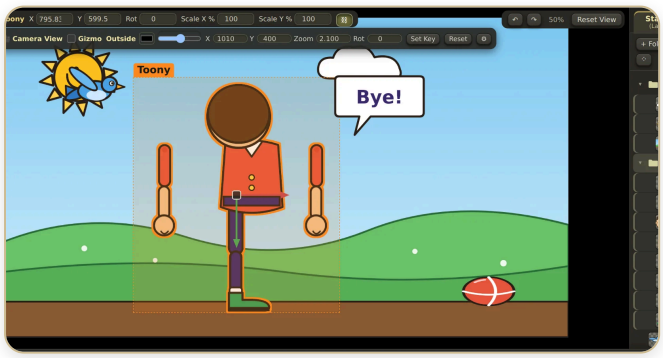
Frame 56 – the arm layer starts to wave.



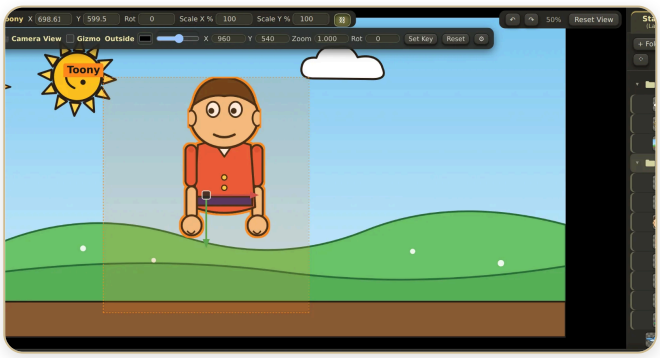
Frame 66 – waves and says Hello!



Frame 84 – Bye! and the head layer turns.



Frame 90 – the back of the head.



Frame 110 – turned around, walking away.

Features used in this project

Feature	Where
Layers with many frames, folders, Switch folder	Steps 2–3
Import Animation (repeat, hold, reversed)	Steps 6–9, 12
Link, move, numeric bar	Steps 4–5
Mouse-wheel frame swaps (mouth, eyes)	Step 7
Hide, Flip	Steps 5, 6, 8, 10, 12
Squash, Stretch	Step 10
Clay	Step 11
Graph Editor	Step 13
Camera and effects	Step 14
Export MP4, clips, Offset	Steps 15–16

Step 0 – Prepare the frames

Make all pictures as **transparent PNG**. Every frame of one layer uses the same canvas size.

Layer	Frames	What the frames show
Legs	8	one walking cycle (side view)
Legs Idle	1	standing legs
Arm R	8	rest, raise, wave
Arm L	2	arm hanging (2 slightly different poses)
Head	8	front (no face) + seven views of the turn
Eyes / Mouth	3 / 4	open-half-closed / closed-small-wide-“O”
Body	1	torso
Bird	8	one flap loop
Ball, Hello, Bye, Cloud, Sun, Background	1 each	props, bubbles and scenery

Step 1 – Project settings

Open File ▸ Project Settings… . Keep the frame at **1920 × 1080**, set **Frame rate 25**, range **0 – 120** and the default key interpolation to **Spline**. Name the project “Toony Says Hello” and save it with `Ctrl+S`.

Step 2 – Build the Staging tree

The finished Staging tree. The number on a row is how many frames the layer holds.

1. Create the folder **Scenery** with three layers, top to bottom: **Cloud**, **Sun**, **Background**.
2. Create the folder **Toony** and add its layers in this order (top = front): **Eyes**, **Mouth**, **Head**, **Arm R**, **Body**, **Arm L**, **Legs Idle**, **Legs**. For each layer open the image dialog and load all its frames together.
3. Create a loose layer **Bird** (8 frames) and a loose layer **Ball**.
4. Create a folder **Bubbles**, right-click ▸ **Switch Folder**, and add layers **Hello** and **Bye** inside it.

> **Tip**
> Name every layer clearly – the side panel is narrow and shortens long names.

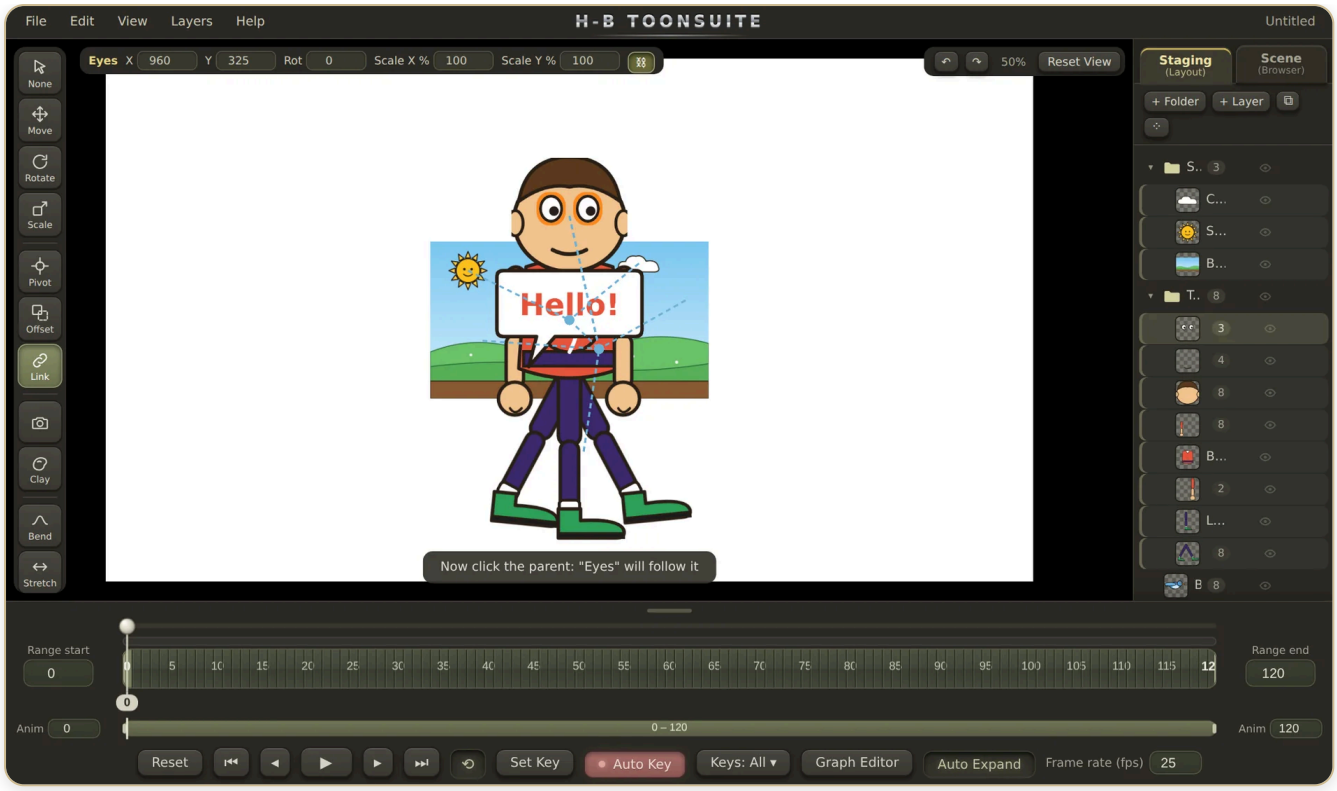
Step 3 – Place everything on the stage

1. Open the **Rig page** of Toony (double-click the folder) and arrange the layers so the head sits on the body, the legs under the belt and the arms at the shoulders. Press **Apply**.
2. Drag **Scenery** onto the frame first, then **Bird**, **Toony**, **Ball** and **Bubbles** (later items appear in front).
3. Scenery arrives small (big folders are scaled down). Select its master and type **Scale X % = 333**; it fills the frame.
4. Numeric bar: Toony master **X = 200**; Ball **X = 1650, Y = 925**; Bubbles **X = 1280, Y = 300**; Bird **Y = 250**.

Everything dropped on the stage (the Scenery folder has not been scaled up yet).

Step 4 – Link the face to the head

Press L. Click **Eyes**, then **Head**. Click **Mouth**, then **Head**. All other layers are already linked to the master. Go back to Move W.



The Link tool waits for the parent after the Eyes layer was clicked.

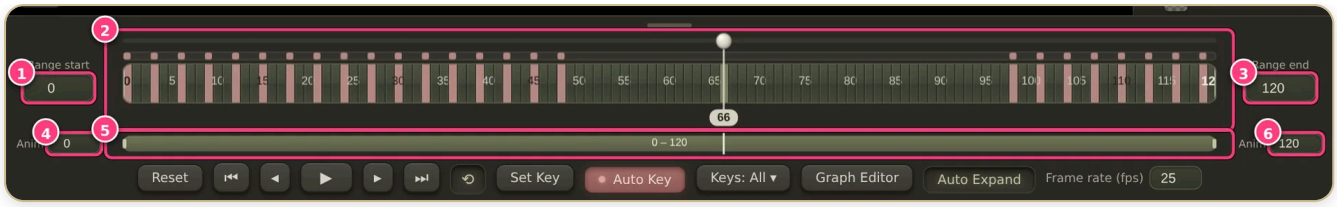
Step 5 – Start with Bubbles and Legs Idle hidden

At frame 0 select **Bubbles** and press `Hide`; then select **Legs Idle** and press `Hide`. (Auto Key is on, so both hides are keyed.)

Step 6 – Walk in (frames 0 – 48) with Import Animation

1. Select **Legs**. In the side panel press **Import Animation…**. Set **Repeat = 2**, **Frames per image = 3**, **Start at frame = 0**, **Load reversed** off. Press **Import**.
2. While you are here, **import the second walk** (for the walk-away) in the same way: Legs again, **Repeat 1, Frames per image 3, Start at frame 98**.
3. Select the **Toony master**: frame 0 **X = 200**, frame 48 **X = 1000**; open the Graph Editor and set both keys to **Linear**.
4. At frame 48 select **Legs** and press `Hide`; select **Legs Idle** and press `Hide` (it appears). At frame 98 select **Legs Idle** and press `Hide` again.

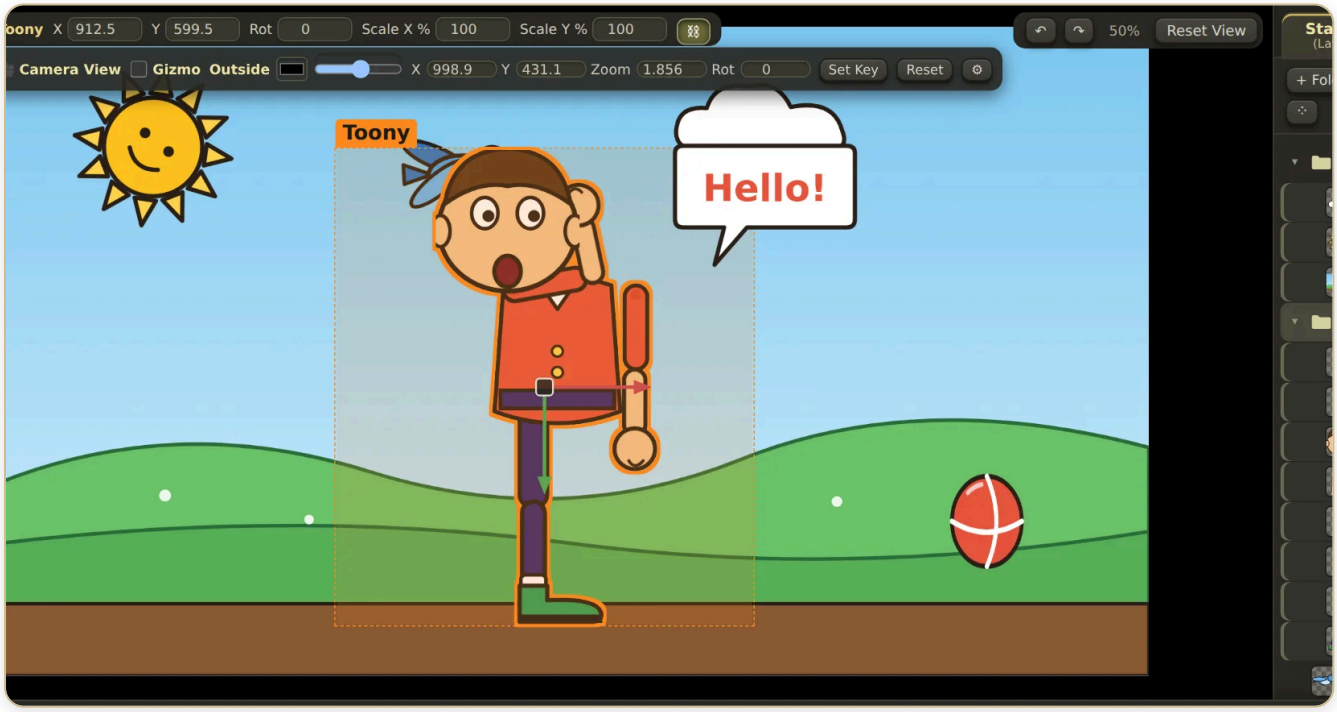
> **Careful**
>
> Do the Import Animation steps **before** the Hide toggles. Imported keys remember the current hidden/visible state, so importing after hiding would carry the hidden state into the new keys.



After the imports the timeline shows the dense stepped keys of the cycles.

Step 7 – Wave and talk (frames 52 – 100)

1. Select **Arm R**. **Import Animation…**: Repeat 1, Frames per image **3**, Start at frame **52**. The arm rises and waves for 24 frames.
2. Import again with **Load reversed** checked, Frames per image **2**, Start at frame **76**: the arm lowers smoothly by itself.
3. Mouth (turn the **mouse wheel** over the mouth at each frame): frame 48 → 1 (closed); 52 → 2; 54 → 3; 57 → 4; 60 → 2; 63 → 3; 66 → 4; 69 → 2; 72 → 3; 75 → 1.
4. Eyes blink: frame 60 → 1 (open), 61 → 2, 62 → 3 (closed), 64 → 2, 65 → 1.
5. Bubbles: frame 52 select Bubbles, press `Hide` (it appears showing **Hello!**); frame 80 hold `Ctrl+Shift` and scroll once so it shows **Bye!**; frame 100 press `Hide` again.



Frame 66: Hello! – the arm frames, the mouth swap and the bubble all work together.

Step 8 – A full head turn

1. Select **Head**. **Import Animation…**: Repeat 1, Frames per image **2**, Start at frame **80**. The head spins through all eight views in 16 frames.

2. At frame 96 turn the wheel over the head until the counter shows **1 / 8** (the front again).
3. At frame 80 select **Eyes** and **Mouth** and press `Hide`; at frame 96 press `Hide` again to bring them back.

Step 9 – The bird: a whole clip in one layer

1. Select **Bird**. **Import Animation…**: **Repeat 7**, Frames per image **2**, Start at frame **0** (112 frames of flapping).
2. Key its **X**: frame 0 **X = 2050**, frame 120 **X = -250**, both **Linear**.

Import Animation for the bird: 8 images, repeated 7 times, 2 frames per image.

Step 10 – The bouncing ball: Squash and Stretch

The ball bounces every 30 frames. For each bounce starting at frame *s* (0, 30, 60, 90) key the following; at frame 120 repeat the contact pose. Move with the numeric bar (Y) and deform with `Squash` / `Stretch`.

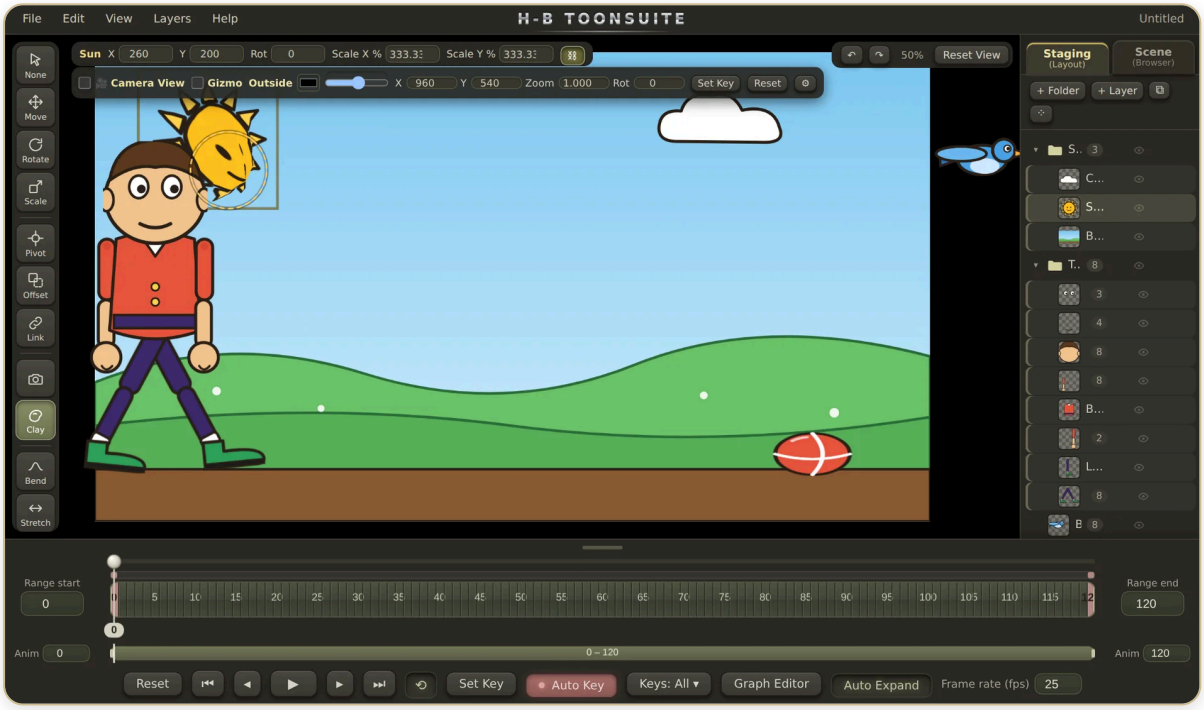
Frame	Y	Shape
s	925	**Squash** – wide (130 %) and flat (72 %)
s + 4	880	**Stretch** – tall (112 %), narrow (90 %)
s + 15	580	Normal round ball (apex)

Frame	Y	Shape
s + 26	880	**Stretch** again before landing

- At frame 108 press **Hide** on the ball – it disappears near the end of the scene.

Step 11 – Sun and cloud: Clay and slow moves

1. At frame 0 select the **Sun**, choose **Clay** and drag a few soft strokes (try `Shift`+drag for a twirl) to make the shape wobbly and friendly.
2. Frame 120: rotate the Sun by **70°** (a key at frame 0 already exists, so it rotates slowly).
3. Frame 120: move the Cloud **300 px to the left**.

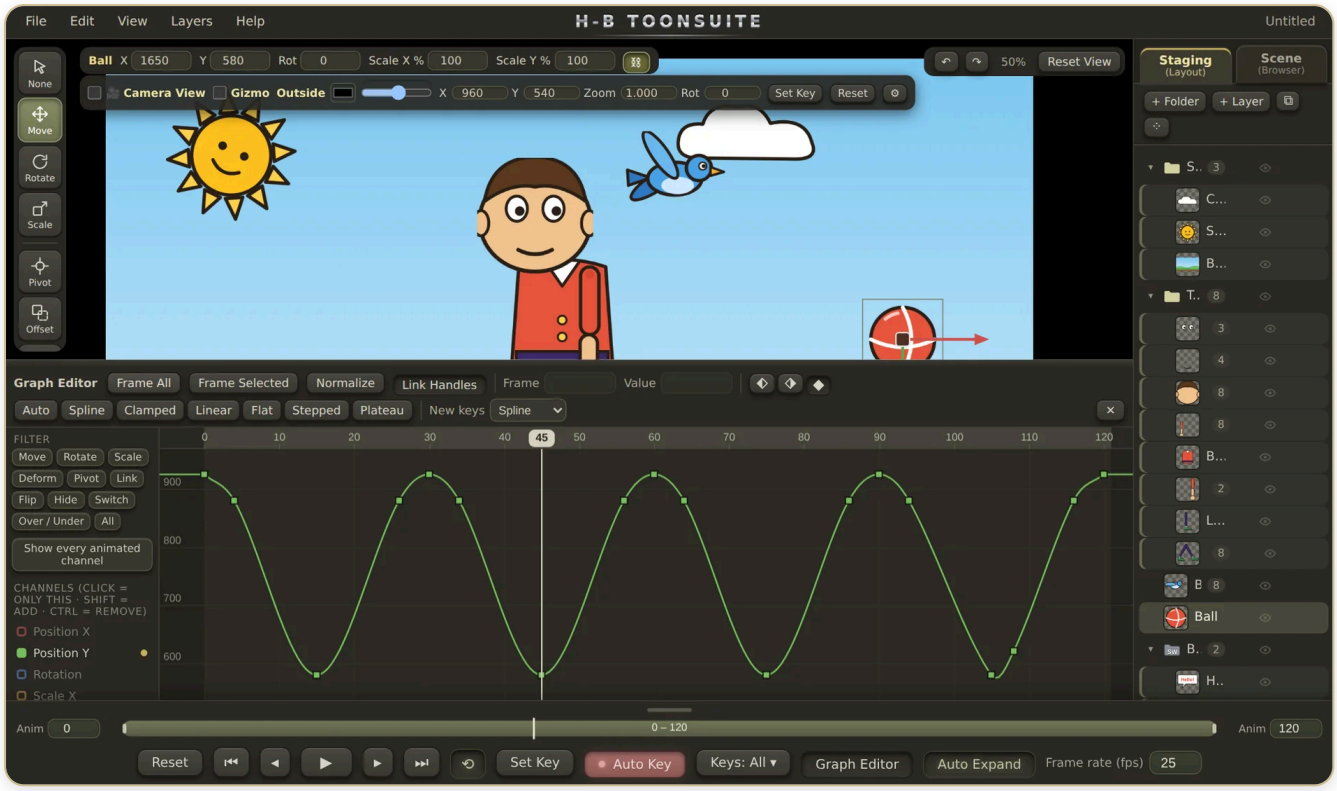


Sculpting the sun with the Clay tool.

Step 12 – Turn around and walk away (frames 98 – 120)

1. Frame 98: select the **Toony master** and press **Flip H**. He now faces left. (The second walk cycle was already imported in step 6.)
2. Key the master **X = 1000** at frame 98 and **X = 650** at frame 120 (Linear).

Step 13 – Polish in the Graph Editor



The Graph Editor showing the bouncing ball's height curve.

- Master and bird **X** keys → **Linear** (steady travel).
- Ball **Y**: **Flat** at the top of each bounce (slow, floaty), **Linear** at the ground (hard hit).
- Press A to see all curves and spot ugly spikes.

Step 14 – The camera

Create a camera (`Layers ▸ New Camera`) and key it with the camera panel:

Frame	X	Y	Zoom
0, 52	960	540	1.0
70 – 92	1010	400	2.1
106, 120	960	540	1.0

In the camera panel switch **Effects on** and set **Vignette = 45**, **Saturation = 12**, **Brightness = 4**.

Frame 84: the camera is zoomed on Toony's face; the head layer is in the middle of its turn.

Step 15 – Export

1. Press `Ctrl+E`.
2. Format **MP4**, Camera **Camera 1**, range **0 – 120** (button **Animation range**), Size **100 %**, FPS **25**, Quality about 80 %.
3. The summary line should read "121 frames ≈ 4.8 s". Click **Start**, wait, and save the file.

Step 16 – Reuse and refine

- Export the walk as a **clip** and the character as a **character file** for your next film (chapter 14).
- Want Toony to lean slightly in one shot? Turn **Auto Key off**, press **Offset** `0` and rotate the body – no key is changed. **Reset Offset** undoes it.
- Save with `Ctrl+S`. Congratulations – you have used the heart of H-B ToonSuite: layers that hold their own frames.

18 Tips, Troubleshooting and Cheat Sheet

Working like a professional

- **Save often**, and turn on *Save Plus* for numbered versions.
- **Name everything.** Clear names make links, clips and characters reusable.
- **Pivot first, animate second.** Fixing a pivot after 100 keys is painful.
- **Pose-to-pose:** key the extreme poses first, then the in-betweens.
- **Hold poses.** Fewer, stronger drawings are the soul of limited animation.
- Test often at 50 % size; export at 100 % only at the end.

Troubleshooting

Problem	Cause and fix
A dropped folder is tiny	The program scales big folders down. Select its master and scale it up in the numeric bar.
An arm rotates around its middle	The pivot is still at the centre. Use the **Pivot** tool P to move it to the shoulder.
The forearm does not follow the arm	Not linked to the upper arm. Use **Link** L: forearm first, upper arm second.
Nothing gets keyed	**Auto Key** is off. Turn it on or use Set Key S.
My change disappears when I play	With Auto Key off on an animated object it became an offset; the keys win. Turn Auto Key on or bake the offset.
The mouth image changes by itself while scrolling the page	The wheel over an object swaps its image. Scroll over empty stage or over the side panel.
White fringe around a picture	Use a transparent PNG, or raise the **Tolerance** of Auto-clean in the layer images dialog.
My cycle jitters	The frames of the layer have different canvas sizes or the part sits in different places. Re-export them on the same canvas.
Import Animation does nothing	The layer needs at least two images and must not be locked.
I cannot select a layer on the stage	It is frozen (snowflake) or hidden behind others. Unfreeze it, use **Spread**, or Alt+click with Clay.
Playback is slow	Turn on **Proxy Images** and use smaller source images.

Problem	Cause and fix
Ctrl+Wheel zooms the browser	Click inside the app first; browser zoom is blocked on this page.
Export fails or is empty	Check the frame range, the chosen camera and keep the tab visible while exporting.

One-page cheat sheet

I want to…	Do this
Make a puppet	Folder of layers → double-click for Rig page → drop on stage → Link → Pivot
Move it across the stage	Key the **master** folder: start and end, Linear
Swap mouths / eyes	Wheel over the part with Auto Key on
Loop a walk	Copy keys of one cycle and paste at its end
Squash a ball	Squash `U` at contact, Stretch `T` in the air
Hide something	Hide with Auto Key on
Turn him around	Flip H on the master
Smooth or snap a motion	Graph Editor `G` → tangent type
Zoom on a face	Camera `C` → drag / wheel → key
Get a video	`Ctrl+E` → MP4 → Start

A Appendix: Default Keyboard Shortcuts

All shortcuts can be changed in **Settings ▸ Shortcuts** `F1`. Defaults:

Playback & keys	Shortcut
Play / Pause	`Space`
Previous / next frame	`Left` `Right`
First / last frame	`Home` `End`
Set key	`S`
Delete keys / objects	`Delete`
Copy / cut / paste keys (mouse over timeline)	`Ctrl+C` `Ctrl+X` `Ctrl+V`
Reverse selected keys	`Shift+R`
Select all	`Ctrl+A`

Tools	Shortcut
None / Move / Rotate / Scale	`Q` `W` `E` `R`
Bend / Stretch / Squash / Skew	`B` `T` `U` `K`
Pivot / Link / Offset (mouse over stage)	`P` `L` `O`
Camera tool on / off (mouse over stage)	`C`
Graph Editor open / close	`G`
Frame all / selected curves	`A` `F`

File, edit, view	Shortcut
New / Open / Save / Save As	`Alt+N` `Ctrl+O` `Ctrl+S` `Ctrl+Shift+S`
Export video / Import vector	`Ctrl+E` `Ctrl+I`
Undo / Redo	`Ctrl+Z` `Ctrl+Y`
Rename layer / folder	`F2`
Duplicate layer	`Ctrl+D`
Zoom app in / out / reset	`Ctrl+=` `Ctrl+-` `Ctrl+0`
Shortcuts settings	`F1`

File, edit, view	Shortcut
Fullscreen	`F11`

Mouse	Action
Wheel over object	Next / previous image
`Ctrl+Wheel` over object	Bring forward / send backward
`Ctrl+Shift+Wheel` over a switch	Next item of the switch folder
`Alt+Wheel` on stage	Zoom the stage view
Middle-drag	Pan stage / timeline / graph
Double-click a folder	Open the Rig page
Double-click a frame	Set a key
Arrow keys / `Shift+Arrows`	Nudge 1 px / 10 px
`Shift` while rotating	Snap to 15°
`Esc`	Cancel a pending link, close panels

— End of the manual —